Engineering Mathematics – III

Unit IV : Complex Integration

Cauchy's Theorems, Series, Singularities, Residues and Contour Integration

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Contents

1 Complex Integration **2**
1.1 Complex Line Integral . . . 2
1.2 Definitions . . . 2
1.3 Cauchy's Integral Theorem . . . 3
1.4 Extension of Cauchy's Integral Theorem . . . 3
1.5 Cauchy's Integral Formula . . . 5
1.6 Summary . . . 6
1.7 Problems . . . 7
1.8 Exercises (with solutions) . . . 9

2 Taylor's and Laurent's Series **11**
2.1 Taylor's Series . . . 11
2.2 Laurent's Series . . . 11
2.3 Problems . . . 12
2.4 Exercises (with solutions) . . . 16

3 Singularities and their Types **19**
3.1 Singular Points . . . 19
3.2 Types of Singularities . . . 19
3.3 Problems . . . 21
3.4 Exercises (with solutions) . . . 23
3.5 Residues . . . 25

4 Cauchy's Residue Theorem **26**
4.1 Cauchy's Residue Theorem . . . 26
4.2 Problems . . . 27
4.3 Exercises (with solutions) . . . 33

5 Evaluation of Real Definite Integrals **35**
5.1 Evaluation of Real Definite Integrals . . . 35
5.2 Type I : $\int_0^{2\pi} f(\cos\theta, \sin\theta)\, d\theta$ . . . 35
5.3 Type II : $\int_{-\infty}^{\infty} \frac{p(x)}{q(x)}\, dx$ . . . 43

1 Complex Integration

Cauchy's Integral Theorem and Cauchy's Integral Formula

1.1 Complex Line Integral

Complex line integral

Let $w = f(z) = u + iv$ be a continuous function of the complex variable $z = x + iy$, and let C be any curve connecting two points a and b. We can express a **complex line integral** in terms of **real line integrals** as follows:

$$\begin{aligned}\int_C f(z)\,dz &= \int_C (u+iv)(dx + i\,dy)\\ &= \int_C (u\,dx - v\,dy) + i\,(u\,dy + v\,dx)\\ \Rightarrow \quad \int_C f(z)\,dz &= \int_C (u\,dx - v\,dy) + i\int_C (u\,dy + v\,dx)\\ &= \text{line integrals of real functions.}\end{aligned}$$

1.2 Definitions

Simple closed curve

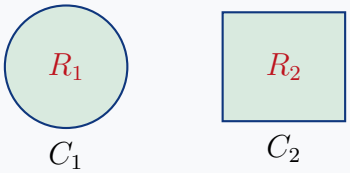
A closed curve which does **not intersect itself**.

Simply connected region

A region R is **simply connected** if every simple closed curve drawn in R encloses points of R only.

Example

Regions bounded by a circle or a square are simply connected regions.



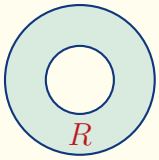
C_1 and C_2 are simple closed curves; R_1 and R_2 are simply connected regions.

Multiply connected region

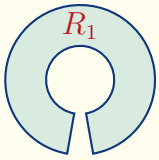
A region which is **not** simply connected is called a **multiply connected region**.

Note

A multiply connected region can be converted into a simply connected region by making a **cross-cut** in between.



R : multiply connected



R_1 : simply connected

1.3 Cauchy's Integral Theorem

Cauchy's Integral Theorem

If $f(z)$ is analytic at every point of the region R bounded by a simple closed curve C, and if $f'(z)$ is continuous throughout this closed region R, then

$$\oint_C f(z)\,dz = 0.$$

Proof: By our earlier discussion, we have

$$\oint_C f(z)\,dz = \oint_C (u\,dx - v\,dy) + i\oint_C (u\,dy + v\,dx)$$

$$= \iint_R \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dx\,dy + i\iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx\,dy$$

$$\left[\because \text{ by Green's theorem, } \quad \oint_C P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy\right]$$

$$= -\iint_R (v_x + u_y)\,dx\,dy + i\iint_R (u_x - v_y)\,dx\,dy$$

$$= 0 + 0 \qquad (\because \text{ by the C–R equations } u_x = v_y \text{ and } u_y = -v_x)$$

$$\oint_C f(z)\,dz = 0.$$

1.4 Extension of Cauchy's Integral Theorem

Cauchy's Integral Theorem for a multiply connected region

Suppose $f(z)$ is analytic in the annular region R bounded by the closed curves C_1 (outer) and C_2 (inner), and on C_1, C_2. Then

$$\oint_{C_1} f(z)\,dz = \oint_{C_2} f(z)\,dz,$$

both integrals being taken in the anticlockwise direction.

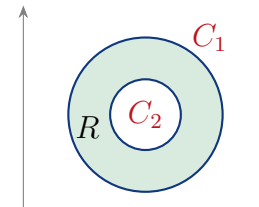


Fig. (a)

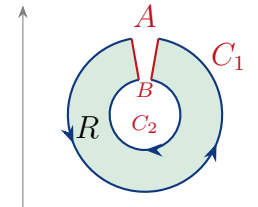


Fig. (b)

Proof: Since R is a multiply connected region (Fig. (a)), we can convert it into a simply connected region by making a simple cross-cut AB (Fig. (b)). In the new simply connected region, bounded by the closed curve $C : C_1 \to AB \to C_2 \to BA$, we have by Cauchy's integral theorem

$$\oint_C f(z)\,dz = 0$$

$$\Rightarrow \quad \left(\int_{C_1} + \int_{AB} + \int_{C_2^*} + \int_{BA}\right) f(z)\,dz = 0$$

where C_2^* denotes C_2 described in the *clockwise* sense (the region must stay on the left). Since $\displaystyle\int_{BA} = -\int_{AB}$,

$$\Rightarrow \quad \left(\int_{C_1} + \int_{AB} - \int_{AB} + \int_{C_2^*}\right) f(z)\,dz = 0$$

$$\Rightarrow \quad \oint_{C_1} f(z)\,dz + \int_{C_2^*} f(z)\,dz = 0$$

$$\Rightarrow \quad \oint_{C_1} f(z)\,dz = -\int_{C_2^*} f(z)\,dz$$

Reversing the direction of C_2^* changes the sign, so

$$\oint_{C_1} f(z)\,dz = \oint_{C_2} f(z)\,dz \qquad \text{(both anticlockwise).}$$

Note

Suppose $f(z)$ is analytic in the region bounded by the exterior contour C and the interior contours $C_1, C_2, \ldots, C_n$ (all described anticlockwise). Then

$$\oint_C f(z)\,dz = \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz + \cdots + \oint_{C_n} f(z)\,dz.$$

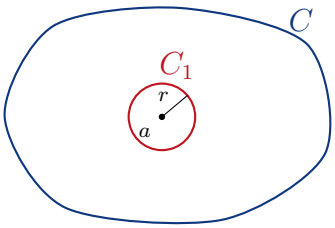
1.5 Cauchy's Integral Formula

Cauchy's Integral Formula

If $f(z)$ is analytic within and on a simple closed curve C of a simply connected region R, and if a is any point lying inside C, then

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a}\, dz.$$

Proof: Given $z = a$ is a point lying inside C. Draw a small circle C_1 with centre at a and radius r, lying inside C.



Then $\dfrac{f(z)}{z-a}$ is analytic in the annular region bounded by C and C_1. Hence, by the extension of Cauchy's integral theorem,

$$\oint_C \frac{f(z)}{z-a}\, dz = \oint_{C_1} \frac{f(z)}{z-a}\, dz. \tag{1}$$

On C_1 we have $|z-a| = r$

$$\Rightarrow \; z - a = re^{i\theta}, \qquad z = a + re^{i\theta}, \qquad dz = re^{i\theta}\, i\, d\theta,$$

and θ varies from 0 to 2π. $\therefore$ (1) $\Rightarrow$

$$\oint_C \frac{f(z)}{z-a}\, dz = \int_0^{2\pi} \frac{f(a+re^{i\theta})}{re^{i\theta}}\, re^{i\theta}\, i\, d\theta$$

$$\Rightarrow \quad \oint_C \frac{f(z)}{z-a}\, dz = i \int_0^{2\pi} f(a+re^{i\theta})\, d\theta.$$

The left-hand side does not depend on r. If we shrink the circle C_1 towards the point a, i.e. if we let $r \to 0$, then (by the continuity of f)

$$\oint_C \frac{f(z)}{z-a}\, dz = i \int_0^{2\pi} f(a)\, d\theta = i\, f(a)\, \big[\theta\big]_0^{2\pi} = 2\pi i\, f(a).$$

$$\oint_C \frac{f(z)}{z-a}\, dz = 2\pi i\, f(a) \qquad \text{(or)} \qquad f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a}\, dz \tag{2}$$

Cauchy's integral formula for derivatives

Differentiating (2) with respect to a (under the integral sign), we get

$$\oint_C \frac{f(z)}{(z-a)^2}\,dz = 2\pi i\, f'(a).$$

Differentiating with respect to a once again,

$$\oint_C \frac{2\,f(z)}{(z-a)^3}\,dz = 2\pi i\, f''(a) \;\Rightarrow\; \oint_C \frac{f(z)}{(z-a)^3}\,dz = \frac{2\pi i}{2!}\, f''(a).$$

Similarly,

$$\oint_C \frac{f(z)}{(z-a)^n}\,dz = \frac{2\pi i}{(n-1)!}\, f^{(n-1)}(a), \qquad n = 1, 2, 3, \ldots$$

1.6 Summary

So far, we have

① If $f(z)$ is analytic inside and on a simple closed curve C, then

$$\oint_C f(z)\,dz = 0 \qquad (\because \text{ by C.I.T.})$$

② If $f(z)$ is analytic inside and on a simple closed curve C and a is a point lying inside C, then

$$\oint_C \frac{f(z)}{z-a}\,dz = 2\pi i\, f(a) \qquad (\because \text{ by C.I.F.})$$

③ Further,

$$\oint_C \frac{f(z)}{(z-a)^2}\,dz = 2\pi i\, f'(a), \qquad \oint_C \frac{f(z)}{(z-a)^3}\,dz = \frac{2\pi i}{2!} f''(a), \;\ldots,$$
$$\oint_C \frac{f(z)}{(z-a)^n}\,dz = \frac{2\pi i}{(n-1)!} f^{(n-1)}(a).$$

Note: point outside C

If a is a point lying **outside** C, then $\dfrac{f(z)}{z-a}$ is analytic inside and on C, and hence by C.I.T.

$$\oint_C \frac{f(z)}{z-a}\,dz = 0.$$

Note: two points a and b

For $\oint_C \dfrac{f(z)}{(z-a)(z-b)}\,dz$ we may use partial fractions,

$$\oint_C \frac{f(z)}{(z-a)(z-b)}\,dz = \oint_C f(z)\left[\frac{A}{z-a}+\frac{B}{z-b}\right]dz,$$

or, when *both* a and b lie inside C, enclose them by small non-overlapping circles C_1 (about a) and C_2 (about b):

$$\oint_C \frac{f(z)}{(z-a)(z-b)}\,dz = \oint_{C_1} \frac{\left(\frac{f(z)}{z-b}\right)}{z-a}\,dz + \oint_{C_2} \frac{\left(\frac{f(z)}{z-a}\right)}{z-b}\,dz.$$

If only one of them, say b, lies inside C, simply write $\oint_C \dfrac{\left(\frac{f(z)}{z-a}\right)}{z-b}\,dz$ and apply C.I.F. once.

1.7 Problems

Problem 1

Evaluate $\oint_C \dfrac{z}{z^2-3z+2}\,dz$, where C is $|z-2|=\frac{1}{2}$.

Solution:

Given

$$I = \oint_C \frac{z}{z^2-3z+2}\,dz = \oint_C \frac{z}{(z-1)(z-2)}\,dz, \qquad C:\ |z-2|=\tfrac{1}{2}.$$

The singular points are $z=1$ and $z=2$.

Put $z=1$**:** $|z-2| = |1-2| = |-1| = 1 > \frac{1}{2}$
$\Rightarrow z=1$ lies **outside** C.

Put $z=2$**:** $|z-2| = |2-2| = 0 < \frac{1}{2}$
$\Rightarrow z=2$ lies **inside** C.

(Geometrically, C is the circle with centre 2 and radius $\frac{1}{2}$, which meets the real axis at 1.5 and 2.5.) Hence

$$I = \oint_C \frac{\left(\frac{z}{z-1}\right)}{z-2}\,dz = \oint_C \frac{g(z)}{z-b}\,dz, \qquad \text{where } g(z)=\frac{z}{z-1},\ b=2.$$

$g(z)$ is analytic inside and on C (its only singularity $z=1$ lies outside C). $\therefore$ by C.I.F.,

$$I = 2\pi i\, g(2) = 2\pi i \left(\frac{z}{z-1}\right)_{z=2} = 2\pi i\left(\frac{2}{2-1}\right)$$

$$I = 4\pi i$$

Problem 2

Evaluate $\oint_C \dfrac{z-3}{z^2+2z+5}\,dz$, where C is $|z+1-i|=2$.

Solution:

To get the factors of z^2+2z+5:

$$z=\frac{-2\pm\sqrt{4-20}}{2}=\frac{-2\pm\sqrt{-16}}{2}=\frac{-2\pm 4i}{2}=-1\pm 2i.$$

$\therefore$

$$I=\oint_C \frac{z-3}{\big(z-(-1+2i)\big)\big(z-(-1-2i)\big)}\,dz.$$

Let $a=-1+2i$ and $b=-1-2i$. To check whether a and b lie inside C, substitute them in $|z+1-i|$:

When $z=a=-1+2i$:

$$\begin{aligned}|z+1-i| &= |-1+2i+1-i|\\ &= |i|=1<2\\ &\Rightarrow a \text{ lies } \textbf{inside } C\end{aligned}$$

When $z=b=-1-2i$:

$$\begin{aligned}|z+1-i| &= |-1-2i+1-i|\\ &= |-3i|=3>2\\ &\Rightarrow b \text{ lies } \textbf{outside } C\end{aligned}$$

Hence

$$I=\oint_C \frac{\left(\dfrac{z-3}{z-(-1-2i)}\right)}{z-(-1+2i)}\,dz=\oint_C \frac{f(z)}{z-a}\,dz,\qquad f(z)=\frac{z-3}{z+1+2i},$$

where $f(z)$ is analytic inside and on C. $\therefore$ by C.I.F.,

$$\begin{aligned}I &= 2\pi i\, f(a)=2\pi i\left(\frac{z-3}{z-(-1-2i)}\right)_{z=-1+2i}\\ &= 2\pi i\left(\frac{-1+2i-3}{-1+2i+1+2i}\right)=2\pi i\left(\frac{2i-4}{4i}\right)=\frac{\pi}{2}(2i-4)\end{aligned}$$

$$I=\pi\,(i-2)$$

Problem 3

Evaluate $\oint_C \dfrac{e^{2z}}{(z+1)^4}\,dz$, where C is $|z|=2$.

Solution:

Given

$$I=\oint_C \frac{e^{2z}}{\big(z-(-1)\big)^4}\,dz=\oint_C \frac{f(z)}{(z-a)^4}\,dz,\qquad f(z)=e^{2z},\ a=-1. \tag{1}$$

Given C is $|z|=2$. Substituting $a=-1$: $|z|=|-1|=1<2\Rightarrow a=-1$ lies inside C.

$\therefore$ (1) $\Rightarrow$ (by C.I.F. for derivatives, with $n=4$)

$$I=\frac{2\pi i}{3!}\,f'''(a)=\frac{2\pi i}{6}\Big(f'''(z)\Big)_{z=-1}.$$

Now $f(z) = e^{2z}$, $f'(z) = 2e^{2z}$, $f''(z) = 4e^{2z}$, $f'''(z) = 8e^{2z}$. Hence

$$I = \frac{\pi i}{3}\left(8e^{2z}\right)_{z=-1} = \frac{\pi i}{3}\left(8e^{-2}\right)$$

$$I = \frac{8}{3}\pi i\, e^{-2}$$

Problem 4

Evaluate $\oint_C \sin z\, dz$, where C is $|z| = 3$.

Solution:

Given $I = \oint_C f(z)\, dz$, where $f(z) = \sin z$ is an analytic function in the region bounded by $C : |z| = 3$ (indeed, everywhere). $\therefore$ by C.I.T.,

$$I = 0$$

1.8 Exercises (with solutions)

Exercise 1

Evaluate $\oint_C \dfrac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)}\, dz$, where C is $|z| = 3$.

Solution:

Let $\phi(z) = \sin \pi z^2 + \cos \pi z^2$ (analytic everywhere). Both $z = 1$ and $z = 2$ lie inside $|z| = 3$. By partial fractions,

$$\frac{1}{(z-1)(z-2)} = \frac{-1}{z-1} + \frac{1}{z-2} \quad \Rightarrow \quad I = -\oint_C \frac{\phi(z)}{z-1}\, dz + \oint_C \frac{\phi(z)}{z-2}\, dz.$$

By C.I.F., $I = 2\pi i\big[-\phi(1) + \phi(2)\big]$, where

$$\phi(1) = \sin \pi + \cos \pi = -1, \qquad \phi(2) = \sin 4\pi + \cos 4\pi = 1.$$

$$I = 2\pi i\,[\,1 + 1\,] = 4\pi i$$

Exercise 2

Evaluate $\oint_C \dfrac{z+1}{z^3 - 2z^2}\, dz$, where C is $|z - 2 - i| = 3$.

Solution:

$z^3 - 2z^2 = z^2(z-2)$, so the singular points are $z = 0$ (order 2) and $z = 2$.

$$z = 0 : \ |0 - 2 - i| = \sqrt{5} \approx 2.24 < 3, \qquad z = 2 : \ |2 - 2 - i| = 1 < 3,$$

so both lie inside C. Enclose them by small circles C_1 (about 0) and C_2 (about 2):

$$I = \oint_{C_1} \frac{g(z)}{z^2}\,dz + \oint_{C_2} \frac{h(z)}{z-2}\,dz, \qquad g(z) = \frac{z+1}{z-2}, \quad h(z) = \frac{z+1}{z^2}.$$

By C.I.F. (and its derivative form),

$$g'(z) = \frac{(z-2)-(z+1)}{(z-2)^2} = \frac{-3}{(z-2)^2} \Rightarrow g'(0) = -\frac{3}{4}, \qquad h(2) = \frac{3}{4}.$$

$$I = 2\pi i\,[g'(0) + h(2)] = 2\pi i\left[-\frac{3}{4} + \frac{3}{4}\right] = 0$$

Exercise 3

Evaluate $\displaystyle\oint_C \frac{\sin^6 z}{\left(z - \frac{\pi}{6}\right)^3}\,dz$, where C is $|z| = 1$.

Solution:

$a = \frac{\pi}{6} \approx 0.52$ lies inside $|z| = 1$. With $f(z) = \sin^6 z$ and $n = 3$,

$$I = \frac{2\pi i}{2!}\, f''\left(\tfrac{\pi}{6}\right) = \pi i\, f''\left(\tfrac{\pi}{6}\right).$$

$f'(z) = 6\sin^5 z\cos z$, $\quad f''(z) = 30\sin^4 z\cos^2 z - 6\sin^6 z$. At $z = \frac{\pi}{6}$ ($\sin = \frac{1}{2}$, $\cos^2 = \frac{3}{4}$):

$$f''\left(\tfrac{\pi}{6}\right) = 30\cdot\frac{1}{16}\cdot\frac{3}{4} - 6\cdot\frac{1}{64} = \frac{90-6}{64} = \frac{21}{16}.$$

$$I = \frac{21\pi i}{16}$$

Exercise 4

Evaluate $\displaystyle\oint_C \frac{z\sec z}{1-z^2}\,dz$, where C is the ellipse $4x^2 + 9y^2 = 9$.

Solution:

$C:\ \dfrac{x^2}{(3/2)^2} + \dfrac{y^2}{1^2} = 1$, an ellipse with semi-axes $\frac{3}{2}$ and 1. The singular points are $z = \pm 1$ (zeros of $1 - z^2$) and the zeros of $\cos z$, $z = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$ Since $\frac{\pi}{2} \approx 1.57 > \frac{3}{2}$, only $z = \pm 1$ lie inside C. By partial fractions,

$$\frac{z}{1-z^2} = -\frac{1}{2}\left[\frac{1}{z-1} + \frac{1}{z+1}\right] \Rightarrow I = -\frac{1}{2}\left[\oint_C \frac{\sec z}{z-1}\,dz + \oint_C \frac{\sec z}{z+1}\,dz\right].$$

Here $\sec z$ is analytic inside and on C, so by C.I.F.

$$I = -\frac{1}{2}\cdot 2\pi i\,\left[\sec 1 + \sec(-1)\right] = -\pi i\,(2\sec 1).$$

$$I = -2\pi i\,\sec 1$$

2 Taylor's and Laurent's Series

Infinite Series in a Complex Plane

We have two types of series in a complex plane:

① Taylor's series

② Laurent's series.

2.1 Taylor's Series

Taylor's Series

Let $f(z)$ be analytic at all points within a circle C_1 with centre at a and radius r. Then at each point z inside C_1, $f(z)$ can be expressed as

$$f(z) = f(a)+(z-a)\,f'(a)+\frac{(z-a)^2}{2!}\,f''(a)+\frac{(z-a)^3}{3!}\,f'''(a)+\cdots=\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}\,(z-a)^n.$$

This series is known as **Taylor's series** of $f(z)$ about $z = a$.

2.2 Laurent's Series

Laurent's Series

Let C_1 and C_2 be two concentric circles with centre a and radii r_1 and r_2 $(r_1 > r_2)$. If $f(z)$ is analytic on C_1, on C_2 and throughout the annular region R between them, then at any point z in R, $f(z)$ can be represented by a convergent series of positive and negative powers of $(z-a)$:

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \cdots + \frac{b_1}{z-a} + \frac{b_2}{(z-a)^2} + \frac{b_3}{(z-a)^3} + \cdots$$

i.e.

$$f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n + \sum_{n=1}^{\infty}\frac{b_n}{(z-a)^n},$$

where

$$a_n = \frac{1}{2\pi i}\oint_{C_1}\frac{f(w)}{(w-a)^{n+1}}\,dw \qquad \text{and} \qquad b_n = \frac{1}{2\pi i}\oint_{C_2}\frac{f(w)}{(w-a)^{-n+1}}\,dw.$$

Notes

① To get Taylor's and Laurent's series, we can use the standard series such as the **binomial**, **logarithmic** and **exponential** series, which get rid of the tedious calculation of a_n and b_n.

② Taylor's series means writing $f(z)$ about $z = a$ in *positive* powers of $(z-a)$.

③ Laurent's series means writing $f(z)$ about $z = a$ in terms of *positive and negative* powers of $(z - a)$.

Standard binomial series used below (valid for $|x| < 1$)

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots = \sum_{n=0}^{\infty} (-1)^n x^n$$

2.3 Problems

Problem 1

For the function $f(z) = \dfrac{z-1}{(z+2)(z+3)}$,

(a) obtain the Taylor's series in the region $|z| < 2$;

(b) obtain the Laurent's series in the region $2 < |z| < 3$.

Solution:

Given

$$\begin{aligned} f(z) &= \frac{z-1}{(z+2)(z+3)} \\ &= \frac{A}{z+2} + \frac{B}{z+3} \\ &= \frac{A(z+3) + B(z+2)}{(z+2)(z+3)} \\ \Rightarrow \quad z - 1 &= A(z+3) + B(z+2). \end{aligned} \tag{1}$$

Put $z = -2$:

$$\begin{aligned} -2 - 1 &= A(-2+3) \\ \Rightarrow \ -3 &= A(1) \\ \Rightarrow \ A &= -3 \end{aligned}$$

Put $z = -3$:

$$\begin{aligned} -3 - 1 &= B(-3+2) \\ \Rightarrow \ -4 &= B(-1) \\ \Rightarrow \ B &= 4 \end{aligned}$$

$\therefore$ From (1),

$$f(z) = \frac{-3}{z+2} + \frac{4}{z+3}. \tag{2}$$

(a) Taylor's series in the region $|z| < 2$

Given $|z| < 2$

$$\Rightarrow \left|\frac{z}{2}\right| < 1, \qquad \text{and obviously} \qquad \left|\frac{z}{3}\right| < 1.$$

$\therefore$ (2) $\Rightarrow$

$$\begin{aligned}
f(z) &= \frac{-3}{2\left(1+\frac{z}{2}\right)} + \frac{4}{3\left(1+\frac{z}{3}\right)} \\
&= -\frac{3}{2}\left(1+\frac{z}{2}\right)^{-1} + \frac{4}{3}\left(1+\frac{z}{3}\right)^{-1} \\
&= -\frac{3}{2}\left[1-\frac{z}{2}+\left(\frac{z}{2}\right)^2-\left(\frac{z}{2}\right)^3+\cdots\right] + \frac{4}{3}\left[1-\frac{z}{3}+\left(\frac{z}{3}\right)^2-\left(\frac{z}{3}\right)^3+\cdots\right] \\
&= -\frac{3}{2}\left[1-\frac{z}{2}+\frac{z^2}{4}-\frac{z^3}{8}+\cdots\right] + \frac{4}{3}\left[1-\frac{z}{3}+\frac{z^2}{9}-\frac{z^3}{27}+\cdots\right] \\
&= \left[-\frac{3}{2}+\frac{4}{3}\right] + z\left[\frac{3}{4}-\frac{4}{9}\right] + z^2\left[-\frac{3}{8}+\frac{4}{27}\right] + z^3\left[\frac{3}{16}-\frac{4}{81}\right] + \cdots \\
&= \left[\frac{-9+8}{6}\right] + z\left[\frac{27-16}{36}\right] + z^2\left[\frac{-81+32}{216}\right] + z^3\left[\frac{243-64}{1296}\right] + \cdots
\end{aligned}$$

$$f(z) = -\frac{1}{6} + \frac{11}{36}z - \frac{49}{216}z^2 + \frac{179}{1296}z^3 - \cdots = \sum_{n=0}^{\infty}(-1)^n\left[\frac{4}{3^{n+1}} - \frac{3}{2^{n+1}}\right]z^n, \qquad |z| < 2$$

is the required Taylor's series of the given $f(z)$.

Check: $f(0) = \dfrac{0-1}{(2)(3)} = -\dfrac{1}{6}$, which agrees with the constant term.

(b) Laurent's series in the region $2 < |z| < 3$

Given $2 < |z| < 3$. Consider

$$\begin{array}{ll}
2 < |z| & |z| < 3 \\
\Rightarrow \dfrac{2}{|z|} < 1 & \Rightarrow \dfrac{|z|}{3} < 1 \\
\Rightarrow \left|\dfrac{2}{z}\right| < 1 & \Rightarrow \left|\dfrac{z}{3}\right| < 1
\end{array}$$

$\therefore$ (2) $\Rightarrow$

$$\begin{aligned}
f(z) &= \frac{-3}{z\left(1+\frac{2}{z}\right)} + \frac{4}{3\left(1+\frac{z}{3}\right)} \\
&= -\frac{3}{z}\left(1+\frac{2}{z}\right)^{-1} + \frac{4}{3}\left(1+\frac{z}{3}\right)^{-1} \\
&= -\frac{3}{z}\left[1-\frac{2}{z}+\left(\frac{2}{z}\right)^2-\left(\frac{2}{z}\right)^3+\cdots\right] + \frac{4}{3}\left[1-\frac{z}{3}+\left(\frac{z}{3}\right)^2-\left(\frac{z}{3}\right)^3+\cdots\right]
\end{aligned}$$

$$= -\frac{3}{z}\left[1 - \frac{2}{z} + \frac{4}{z^2} - \frac{8}{z^3} + \cdots\right] + \frac{4}{3}\left[1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \cdots\right]$$
$$= \left[-\frac{3}{z} + \frac{6}{z^2} - \frac{12}{z^3} + \frac{24}{z^4} - \cdots\right] + \left[\frac{4}{3} - \frac{4z}{9} + \frac{4z^2}{27} - \frac{4z^3}{81} + \cdots\right]$$

$$f(z) = \cdots + \frac{24}{z^4} - \frac{12}{z^3} + \frac{6}{z^2} - \frac{3}{z} + \frac{4}{3} - \frac{4}{9}z + \frac{4}{27}z^2 - \frac{4}{81}z^3 + \cdots, \qquad 2 < |z| < 3$$
$$\text{i.e.} \quad f(z) = \sum_{n=1}^{\infty} \frac{(-1)^n\, 3 \cdot 2^{n-1}}{z^n} + \sum_{n=0}^{\infty} \frac{(-1)^n\, 4}{3^{n+1}}\, z^n$$

is the required Laurent's series about $z = 0$.

Problem 2

Obtain the Laurent's series expansion of $\dfrac{1}{(z+1)(z+3)}$ in powers of $(z+1)$ in the region $0 < |z+1| < 2$.

Solution:

Given

$$f(z) = \frac{1}{(z+1)(z+3)}.$$

To get the Laurent's series in powers of $(z+1)$, i.e. in $0 < |z+1| < 2$, let $u = z+1$, so that $z+3 = u+2$. Then

$$f(z) = \frac{1}{u(u+2)}$$
$$= \frac{A}{u} + \frac{B}{u+2}$$
$$\Rightarrow \quad 1 = A(u+2) + Bu.$$

Put $u = 0$:

$$1 = 2A$$
$$\Rightarrow \; A = \tfrac{1}{2}$$

Put $u = -2$:

$$1 = -2B$$
$$\Rightarrow \; B = -\tfrac{1}{2}$$

$$\Rightarrow \quad f(z) = \frac{A}{u} + \frac{B}{u+2} = \frac{1}{2u} - \frac{1}{2} \cdot \frac{1}{u+2}. \tag{1}$$

Now the region is $0 < |z+1| < 2$

$$\Rightarrow \; 0 < |u| < 2$$
$$\Rightarrow \; 0 < |u| \quad \text{and} \quad |u| < 2$$
$$\Rightarrow \quad \frac{|u|}{2} < 1$$
$$\Rightarrow \quad \left|\frac{u}{2}\right| < 1.$$

$\therefore$ (1) $\Rightarrow$

$$f(z) = \frac{1}{2u} - \frac{1}{2} \cdot \frac{1}{2\left(1 + \frac{u}{2}\right)}$$

$$= \frac{1}{2u} - \frac{1}{4}\left(1 + \frac{u}{2}\right)^{-1}$$
$$= \frac{1}{2u} - \frac{1}{4}\left[1 - \frac{u}{2} + \left(\frac{u}{2}\right)^2 - \left(\frac{u}{2}\right)^3 + \cdots\right]$$
$$= \frac{1}{2u} - \frac{1}{4} + \frac{u}{8} - \frac{u^2}{16} + \frac{u^3}{32} - \cdots$$

Put $u = z + 1$:

$$f(z) = \frac{1}{2(z+1)} - \frac{1}{4}\left[1 - \frac{z+1}{2} + \frac{(z+1)^2}{4} - \frac{(z+1)^3}{8} + \cdots\right]$$

$$\text{i.e.} \quad f(z) = \frac{1}{2(z+1)} - \frac{1}{4} + \frac{z+1}{8} - \frac{(z+1)^2}{16} + \cdots$$

$$\text{i.e.} \quad f(z) = \frac{1}{2(z+1)} + \sum_{n=0}^{\infty} \frac{(-1)^{n+1}(z+1)^n}{2^{n+2}}, \quad 0 < |z+1| < 2$$

is the required Laurent's series of $f(z)$ in powers of $(z + 1)$.

2.4 Exercises (with solutions)

Exercise 1

Expand $\dfrac{1}{z^2-3z+2}$ in terms of (a) Taylor's series in $|z|<1$; (b) Laurent's series in $1<|z|<2$.

Solution:

$$\begin{aligned} f(z) &= \frac{1}{z^2-3z+2} = \frac{1}{(z-1)(z-2)} \\ &= \frac{A}{z-1} + \frac{B}{z-2} \\ \Rightarrow \quad 1 &= A(z-2) + B(z-1). \end{aligned}$$

Put $z=1$: $1 = A(-1) \Rightarrow A = -1$ | **Put** $z=2$: $1 = B(1) \Rightarrow B = 1$

$$\therefore \quad f(z) = \frac{-1}{z-1} + \frac{1}{z-2} = \frac{1}{1-z} - \frac{1}{2-z}. \tag{1}$$

(a) Taylor's series in $|z|<1$

$|z|<1 \Rightarrow |z|<1$ and obviously $\left|\frac{z}{2}\right|<1$. $\therefore$ (1) $\Rightarrow$

$$\begin{aligned} f(z) &= (1-z)^{-1} - \frac{1}{2\left(1-\frac{z}{2}\right)} = (1-z)^{-1} - \frac{1}{2}\left(1-\frac{z}{2}\right)^{-1} \\ &= \left[1+z+z^2+z^3+\cdots\right] - \frac{1}{2}\left[1+\frac{z}{2}+\frac{z^2}{4}+\frac{z^3}{8}+\cdots\right] \\ &= \left(1-\frac{1}{2}\right) + z\left(1-\frac{1}{4}\right) + z^2\left(1-\frac{1}{8}\right) + z^3\left(1-\frac{1}{16}\right) + \cdots \end{aligned}$$

$$f(z) = \frac{1}{2} + \frac{3}{4}z + \frac{7}{8}z^2 + \frac{15}{16}z^3 + \cdots = \sum_{n=0}^{\infty}\left(1-\frac{1}{2^{n+1}}\right)z^n, \qquad |z|<1$$

(b) Laurent's series in $1<|z|<2$

$1<|z| \Rightarrow \left|\frac{1}{z}\right|<1$ and $|z|<2 \Rightarrow \left|\frac{z}{2}\right|<1$. $\therefore$ (1) $\Rightarrow$

$$\begin{aligned} f(z) &= \frac{-1}{z\left(1-\frac{1}{z}\right)} - \frac{1}{2\left(1-\frac{z}{2}\right)} \\ &= -\frac{1}{z}\left(1-\frac{1}{z}\right)^{-1} - \frac{1}{2}\left(1-\frac{z}{2}\right)^{-1} \\ &= -\frac{1}{z}\left[1+\frac{1}{z}+\frac{1}{z^2}+\cdots\right] - \frac{1}{2}\left[1+\frac{z}{2}+\frac{z^2}{4}+\cdots\right] \end{aligned}$$

$$f(z) = \cdots - \frac{1}{z^3} - \frac{1}{z^2} - \frac{1}{z} - \frac{1}{2} - \frac{z}{4} - \frac{z^2}{8} - \cdots = -\sum_{n=1}^{\infty} \frac{1}{z^n} - \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}, \qquad 1 < |z| < 2$$

Exercise 2

Expand the function $f(z) = \dfrac{e^z}{(z-1)^2}$ in terms of Laurent's series about $z = 1$.

Solution:

Let $u = z - 1$, so $z = 1 + u$. Then

$$\begin{aligned} f(z) &= \frac{e^{1+u}}{u^2} = \frac{e \cdot e^u}{u^2} \\ &= \frac{e}{u^2}\left[1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \frac{u^4}{4!} + \cdots\right] \\ &= e\left[\frac{1}{u^2} + \frac{1}{u} + \frac{1}{2!} + \frac{u}{3!} + \frac{u^2}{4!} + \cdots\right]. \end{aligned}$$

Put $u = z - 1$:

$$f(z) = e\left[\frac{1}{(z-1)^2} + \frac{1}{z-1} + \frac{1}{2} + \frac{z-1}{6} + \frac{(z-1)^2}{24} + \cdots\right]$$

$$\text{i.e.} \quad f(z) = e\sum_{n=0}^{\infty} \frac{(z-1)^{n-2}}{n!}, \qquad 0 < |z-1| < \infty$$

Exercise 3

Obtain the Laurent's series expansions of $\dfrac{(z-2)(z+2)}{(z+1)(z+4)}$ which are valid in (i) $1 < |z| < 4$; (ii) $|z| > 4$.

Solution:

Here the degrees of the numerator and the denominator are equal, so we first divide:

$$\begin{aligned} f(z) &= \frac{z^2 - 4}{z^2 + 5z + 4} = 1 + \frac{-5z - 8}{(z+1)(z+4)} \\ &= 1 + \frac{A}{z+1} + \frac{B}{z+4} \\ \Rightarrow \quad -5z - 8 &= A(z+4) + B(z+1). \end{aligned}$$

Put $z = -1$: $-3 = 3A \Rightarrow A = -1$ | **Put** $z = -4$: $12 = -3B \Rightarrow B = -4$

$$\therefore \quad f(z) = 1 - \frac{1}{z+1} - \frac{4}{z+4}. \tag{1}$$

(i) Laurent's series in $1 < |z| < 4$

$1 < |z| \Rightarrow \left|\frac{1}{z}\right| < 1$ and $|z| < 4 \Rightarrow \left|\frac{z}{4}\right| < 1$. $\therefore$ (1) $\Rightarrow$

$$\begin{aligned} f(z) &= 1 - \frac{1}{z\left(1+\frac{1}{z}\right)} - \frac{4}{4\left(1+\frac{z}{4}\right)} \\ &= 1 - \frac{1}{z}\left(1+\frac{1}{z}\right)^{-1} - \left(1+\frac{z}{4}\right)^{-1} \\ &= 1 - \frac{1}{z}\left[1 - \frac{1}{z} + \frac{1}{z^2} - \cdots\right] - \left[1 - \frac{z}{4} + \frac{z^2}{16} - \frac{z^3}{64} + \cdots\right] \\ &= 1 - \left[\frac{1}{z} - \frac{1}{z^2} + \frac{1}{z^3} - \cdots\right] - 1 + \frac{z}{4} - \frac{z^2}{16} + \frac{z^3}{64} - \cdots \end{aligned}$$

$$f(z) = \cdots - \frac{1}{z^3} + \frac{1}{z^2} - \frac{1}{z} + \frac{z}{4} - \frac{z^2}{16} + \frac{z^3}{64} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^n}{z^n} + \sum_{n=1}^{\infty} \frac{(-1)^{n+1} z^n}{4^n}, \qquad 1 < |z| < 4$$

(ii) Laurent's series in $|z| > 4$

$|z| > 4 \Rightarrow \left|\frac{4}{z}\right| < 1$, and hence also $\left|\frac{1}{z}\right| < 1$. $\therefore$ (1) $\Rightarrow$

$$\begin{aligned} f(z) &= 1 - \frac{1}{z}\left(1+\frac{1}{z}\right)^{-1} - \frac{4}{z}\left(1+\frac{4}{z}\right)^{-1} \\ &= 1 - \frac{1}{z}\left[1 - \frac{1}{z} + \frac{1}{z^2} - \cdots\right] - \frac{4}{z}\left[1 - \frac{4}{z} + \frac{16}{z^2} - \cdots\right] \\ &= 1 - \left[\frac{1}{z} - \frac{1}{z^2} + \frac{1}{z^3} - \cdots\right] - \left[\frac{4}{z} - \frac{16}{z^2} + \frac{64}{z^3} - \cdots\right] \end{aligned}$$

$$f(z) = 1 - \frac{5}{z} + \frac{17}{z^2} - \frac{65}{z^3} + \cdots = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n \left(1 + 4^n\right)}{z^n}, \qquad |z| > 4$$

3 Singularities and their Types

Isolated Singularities, Poles and Residues

3.1 Singular Points

Singular point

The point $z = a$ is said to be a **singular point** (or **singularity**) of $f(z)$ if $f(z)$ is not analytic at $z = a$.

Isolated singular point

A point $z = a$ is said to be an **isolated singular point** of $f(z)$ if

(i) $f(z)$ is not analytic at $z = a$, and

(ii) $f(z)$ is analytic in some deleted neighbourhood $0 < |z - a| < r$ of $z = a$ (i.e. there is no other singular point near $z = a$).

Example

If $f(z) = \dfrac{1}{z}$, then $z = 0$ is an isolated singular point, since $f(z)$ is analytic everywhere except at $z = 0$.

Note: Regular part and principal part

If $z = a$ is an isolated singular point of $f(z)$, the Laurent series expansion of $f(z)$ about $z = a$ can be written as

$$f(z) = \underbrace{\sum_{n=0}^{\infty} a_n (z-a)^n}_{\text{analytic (or regular) part}} + \underbrace{\sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n}}_{\text{principal part}}.$$

The type of the singularity is decided by the **principal part**.

3.2 Types of Singularities

1. Essential Singularity

Essential singularity

An isolated singular point $z = a$ is said to be an **essential singularity** of $f(z)$ if the Laurent series of $f(z)$ about $z = a$ possesses an **infinite number of terms** in the principal part.

Example: $f(z) = e^{1/z}$ at $z = 0$

$$f(z) = e^{1/z} = 1 + \frac{(1/z)}{1!} + \frac{(1/z)^2}{2!} + \frac{(1/z)^3}{3!} + \cdots$$
$$= 1 + \frac{1}{(z-0)} + \frac{1}{2!}\frac{1}{(z-0)^2} + \frac{1}{3!}\frac{1}{(z-0)^3} + \cdots$$

This contains an infinite number of terms of the principal part about $z = 0$.
$\Rightarrow z = 0$ **is an essential singular point.**

2. Removable Singularity

Removable singularity

An isolated singular point $z = a$ of $f(z)$ is said to be a **removable singularity** if the Laurent series of $f(z)$ about $z = a$ **does not contain the principal part**.

Example: $f(z) = \dfrac{\sin z}{z}$ at $z = 0$

$$f(z) = \frac{1}{z}\sin z = \frac{1}{z}\left[z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots\right]$$
$$= 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots$$
$$= 1 - \frac{(z-0)^2}{3!} + \frac{(z-0)^4}{5!} - \cdots$$

There are no terms of the principal part.
$\Rightarrow z = 0$ **is a removable singular point of the given** $f(z)$**.**

3. Poles

Pole

A **pole** of a function $f(z)$ at $z = a$ is an isolated singularity where the Laurent series expansion of $f(z)$ about $z = a$ contains only a **finite number of terms** in the principal part (i.e. the negative powers of $(z - a)$).
In other words, if the Laurent series of $f(z)$ about $z = a$ is

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{N} \frac{b_n}{(z-a)^n}, \qquad b_N \neq 0,$$

where N is a finite positive integer, then $z = a$ is a **pole of order** N.

- **Order of the pole:** the *highest* power of $\dfrac{1}{z-a}$ present in the principal part is called the order of the pole. If the principal part contains terms up to $\dfrac{1}{(z-a)^N}$, then $z = a$ is a pole

of order N.

- **Simple pole:** a pole of order 1 (i.e. the principal part contains only $\frac{b_1}{z-a}$) is called a **simple pole**.

Pole: test by limit

A pole of $f(z)$ at $z = a$ can also be described by the behaviour of $f(z)$ as $z \to a$. For a **pole of order** N at $z = a$, the following limit exists and is finite:

$$\lim_{z\to a}(z-a)^N f(z) = L, \qquad \text{where } L \neq 0.$$

For a **simple pole** (order 1):

$$\lim_{z\to a}(z-a) f(z) = L, \qquad \text{where } L \neq 0.$$

Example

For $f(z) = \dfrac{z-1}{(z-2)(z-3)^4}$, we have

- $z = 2$ is a pole of order 1 (simple pole);
- $z = 3$ is a pole of order 4.

Summary: nature of an isolated singularity $z = a$

Principal part of the Laurent series about $z = a$	**Nature of** $z = a$
No terms	Removable singularity
Finitely many terms, highest $\dfrac{1}{(z-a)^N}$	Pole of order N
Infinitely many terms	Essential singularity

3.3 Problems

Problem

Find the nature of the singularity of the following functions:

① $f(z) = \dfrac{\sin z - z}{z^3}$

② $f(z) = \sin\left(\dfrac{1}{z+1}\right)$

③ $f(z) = \dfrac{e^{2z}}{(z-1)^4}$

① $f(z) = \dfrac{\sin z - z}{z^3}$

Solution:

Clearly $z = 0$ is an isolated singular point. Further,

$$\begin{aligned} f(z) = \frac{\sin z - z}{z^3} &= \frac{1}{z^3}\left[\left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots\right) - z\right] \\ &= \frac{1}{z^3}\left[-\frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots\right] \\ &= -\frac{1}{3!} + \frac{z^2}{5!} - \frac{z^4}{7!} + \cdots \end{aligned}$$

This contains no principal part.

$\Rightarrow$ $z = 0$ is a **removable singular point**.

② $f(z) = \sin\left(\dfrac{1}{z+1}\right)$

Solution:

Clearly $z = -1$ is an isolated singular point. Further,

$$\begin{aligned} f(z) = \sin\left(\frac{1}{z+1}\right) &= \frac{1}{z+1} - \frac{\left(\frac{1}{z+1}\right)^3}{3!} + \frac{\left(\frac{1}{z+1}\right)^5}{5!} - \cdots \\ \Rightarrow \quad f(z) &= \frac{1}{z+1} - \frac{1}{3!}\frac{1}{(z+1)^3} + \frac{1}{5!}\frac{1}{(z+1)^5} - \cdots \end{aligned}$$

This contains an infinite number of terms of the principal part.

$\Rightarrow$ $z = -1$ is an **essential singular point**.

③ $f(z) = \dfrac{e^{2z}}{(z-1)^4}$

Solution:

Clearly $z = 1$ is an isolated singular point. Also

$$\lim_{z\to 1}(z-1)^4 f(z) = \lim_{z\to 1}(z-1)^4 \frac{e^{2z}}{(z-1)^4} = \lim_{z\to 1} e^{2z} = e^2 \neq 0.$$

$\Rightarrow$ $z = 1$ is a **pole of order 4**.

3.4 Exercises (with solutions)

Exercise

Identify the nature of the singular points of the functions

① $\dfrac{\tan z}{z}$

② $z\,e^{1/z^2}$

③ $\dfrac{z-2}{(z-3)^2(z-1)}$

① $f(z) = \dfrac{\tan z}{z} = \dfrac{\sin z}{z\cos z}$

Solution:

The singular points are $z = 0$ and the zeros of $\cos z$, i.e. $z = (2k+1)\dfrac{\pi}{2}$, $k \in \mathbb{Z}$. Each is isolated.

At $z = 0$:

$$f(z) = \frac{1}{z}\left[z + \frac{z^3}{3} + \frac{2z^5}{15} + \cdots\right] = 1 + \frac{z^2}{3} + \frac{2z^4}{15} + \cdots$$

No principal part $\Rightarrow z = 0$ is a **removable singularity** (indeed $\lim_{z\to 0}\frac{\tan z}{z} = 1$).

At $z = a = (2k+1)\frac{\pi}{2}$: $\cos a = 0$, $\sin a = \pm 1 \neq 0$ and $a \neq 0$, so

$$\lim_{z\to a}(z-a)f(z) = \lim_{z\to a}\frac{\sin z}{z}\cdot\frac{z-a}{\cos z} = \frac{\sin a}{a}\cdot\frac{1}{-\sin a} = -\frac{1}{a} \neq 0 \qquad \text{(L'Hospital's rule).}$$

$z = 0$ is a removable singularity; $z = (2k+1)\dfrac{\pi}{2}$ $(k \in \mathbb{Z})$ are **simple poles**.

② $f(z) = z\,e^{1/z^2}$

Solution:

$z = 0$ is the only (isolated) singular point.

$$f(z) = z\left[1 + \frac{1}{z^2} + \frac{1}{2!\,z^4} + \frac{1}{3!\,z^6} + \cdots\right] = z + \frac{1}{z} + \frac{1}{2!\,z^3} + \frac{1}{3!\,z^5} + \cdots$$

Infinitely many terms in the principal part.

$\Rightarrow$ $z = 0$ is an **essential singular point**.

③ $f(z) = \dfrac{z-2}{(z-3)^2(z-1)}$

Solution:

The singular points are $z = 3$ and $z = 1$.

$$\lim_{z\to 3}(z-3)^2 f(z) = \lim_{z\to 3}\frac{z-2}{z-1} = \frac{1}{2} \neq 0 \quad \Rightarrow \; z = 3 \text{ is a pole of order 2,}$$

$$\lim_{z\to 1}(z-1) f(z) = \lim_{z\to 1}\frac{z-2}{(z-3)^2} = \frac{-1}{4} \neq 0 \quad \Rightarrow \; z = 1 \text{ is a simple pole.}$$

$z = 3$ is a **pole of order 2**; $z = 1$ is a **simple pole**.

3.5 Residues

Residue

The **residue** of a function $f(z)$ at an isolated singular point $z = a$ is the coefficient of $\dfrac{1}{z-a}$ (i.e. b_1) in the Laurent series expansion of $f(z)$ about $z = a$.

Evaluation of Residues

For poles, the evaluation of residues can be done as follows.

(1) Simple pole

If $f(z)$ has a simple pole at $z = a$, then the residue of $f(z)$ at $z = a$ is given by

$$\Big[\operatorname{Res} f(z)\Big]_{z=a} = \lim_{z\to a}(z-a)\,f(z).$$

(2) Pole of order n

If $z = a$ is a pole of order n, then

$$\Big[\operatorname{Res} f(z)\Big]_{z=a} = \frac{1}{(n-1)!}\lim_{z\to a}\frac{d^{n-1}}{dz^{n-1}}\Big[(z-a)^n f(z)\Big].$$

(3) Simple pole of $f(z) = \phi(z)/\psi(z)$

Suppose $z = a$ is a pole of order 1 and $f(z) = \dfrac{\phi(z)}{\psi(z)}$, where $\phi(a) \neq 0$, $\psi(a) = 0$ and $\psi'(a) \neq 0$. Then

$$\Big[\operatorname{Res} f(z)\Big]_{z=a} = \frac{\phi(a)}{\psi'(a)}.$$

Note

For an essential singularity none of the above limit formulae applies; the residue must be read off directly as the coefficient of $\dfrac{1}{z-a}$ in the Laurent series. For example, from the expansion of $e^{1/z}$ above, $\Big[\operatorname{Res} e^{1/z}\Big]_{z=0} = 1$, and from Problem ②, $\Big[\operatorname{Res} \sin\frac{1}{z+1}\Big]_{z=-1} = 1$.

4 Cauchy's Residue Theorem

Evaluation of Contour Integrals: Worked Problems

4.1 Cauchy's Residue Theorem

Cauchy's Residue Theorem

If $f(z)$ is analytic at all points inside and on a simple closed curve C, except at a finite number of isolated singular points $z_1, z_2, \ldots, z_n$ lying inside C, then

$$\oint_C f(z)\,dz = 2\pi i \left[\text{Sum of the residues of } f(z) \text{ at } z_1, z_2, \ldots, z_n\right]$$
$$= 2\pi i \sum_{k=1}^{n} \Big[\operatorname{Res} f(z)\Big]_{z=z_k}.$$

Residue formulae used below

(1) Simple pole at $z = a$: $\Big[\operatorname{Res} f(z)\Big]_{z=a} = \lim_{z\to a}(z-a)\,f(z)$.

(2) Pole of order n at $z = a$: $\Big[\operatorname{Res} f(z)\Big]_{z=a} = \dfrac{1}{(n-1)!}\lim_{z\to a}\dfrac{d^{n-1}}{dz^{n-1}}\Big[(z-a)^n f(z)\Big]$.

(3) Simple pole of $f = \phi/\psi$ with $\phi(a) \neq 0$, $\psi(a) = 0$, $\psi'(a) \neq 0$:

$$\Big[\operatorname{Res} f(z)\Big]_{z=a} = \frac{\phi(a)}{\psi'(a)}.$$

(4) Any isolated singularity (in particular an essential one): $\Big[\operatorname{Res} f(z)\Big]_{z=a} = b_1$, the coefficient of $\dfrac{1}{z-a}$ in the Laurent series of $f(z)$ about $z = a$.

Procedure

Step 1. Write $I = \oint_C f(z)\,dz$ and identify $f(z)$ and C.

Step 2. Find the poles of $f(z)$ by equating the denominator to zero, and their orders.

Step 3. Check which poles lie inside C (substitute each pole into the equation of C).

Step 4. Find the residue at each pole inside C.

Step 5. $I = 2\pi i \times$ (sum of these residues).

4.2 Problems

Evaluate the following integrals using Cauchy's residue theorem.

Problem 1

Evaluate $\oint_C \dfrac{e^{2z}\,dz}{(z+2)(z+1)^2}$, where C is $|z|=3$.

Solution:

Given

$$I = \oint_C \frac{e^{2z}\,dz}{(z+2)(z+1)^2} = \oint_C f(z)\,dz, \qquad \text{where } f(z) = \frac{e^{2z}}{(z+2)(z+1)^2} \text{ and } C \text{ is } |z|=3.$$

By Cauchy's residue theorem,

$$I = 2\pi i \,\big(\text{Sum of the residues of } f(z) \text{ at its isolated singular points inside } C\big). \tag{1}$$

Poles. The poles (isolated singular points) are given by equating the denominator to zero:

$$(z+2)(z+1)^2 = 0 \quad \Rightarrow \quad z = -2,\ -1,\ -1.$$

$\Rightarrow z=-2$ is a simple pole and $z=-1$ is a pole of order 2.

Location. C is $|z|=3$.

$$\begin{aligned} &\text{When } z=-2: \quad |z| = |-2| = 2 < 3 \ \Rightarrow\ z=-2 \text{ lies inside } C.\\ &\text{When } z=-1: \quad |z| = |-1| = 1 < 3 \ \Rightarrow\ z=-1 \text{ lies inside } C. \end{aligned}$$

$\Rightarrow$ Both $z=-2$ and $z=-1$ lie inside C.

Residue at $z=-2$ (simple pole).

$$\begin{aligned} \Big[\operatorname{Res} f(z)\Big]_{z=-2} &= \lim_{z\to -2}\big(z-(-2)\big)f(z) = \lim_{z\to -2}(z+2)\,\frac{e^{2z}}{(z+2)(z+1)^2}\\ &= \lim_{z\to -2}\frac{e^{2z}}{(z+1)^2} = \frac{e^{2(-2)}}{(-2+1)^2} = \frac{e^{-4}}{(-1)^2} = \frac{e^{-4}}{1} \end{aligned}$$

$$\Big[\operatorname{Res} f(z)\Big]_{z=-2} = e^{-4} \tag{i}$$

Residue at $z=-1$ (pole of order 2). Using formula (2) with $n=2$:

$$\begin{aligned} \Big[\operatorname{Res} f(z)\Big]_{z=-1} &= \frac{1}{(2-1)!}\lim_{z\to -1}\frac{d^{2-1}}{dz^{2-1}}\Big[\big(z-(-1)\big)^2 f(z)\Big]\\ &= \frac{1}{1!}\lim_{z\to -1}\frac{d}{dz}\left[(z+1)^2\,\frac{e^{2z}}{(z+2)(z+1)^2}\right]\\ &= \lim_{z\to -1}\frac{d}{dz}\left[\frac{e^{2z}}{z+2}\right]\\ &= \lim_{z\to -1}\left[\frac{(z+2)\cdot 2e^{2z} - e^{2z}\cdot(1)}{(z+2)^2}\right] \qquad \left(\text{quotient rule: } \frac{d}{dz}\frac{u}{v} = \frac{vu'-uv'}{v^2}\right)\\ &= \frac{(-1+2)\cdot 2e^{-2} - e^{-2}}{(-1+2)^2} = \frac{(1)\,2e^{-2} - e^{-2}}{(1)^2} = \frac{2e^{-2}-e^{-2}}{1} \end{aligned}$$

$$\left[\operatorname{Res} f(z)\right]_{z=-1} = e^{-2} \qquad \text{(ii)}$$

Result. Substituting (i) and (ii) in (1):

$$\begin{aligned} I &= 2\pi i \left[\left[\operatorname{Res} f(z)\right]_{z=-2} + \left[\operatorname{Res} f(z)\right]_{z=-1}\right] \\ &= 2\pi i \left[e^{-4} + e^{-2}\right] = 2\pi i \left[\frac{1}{e^4} + \frac{1}{e^2}\right] = 2\pi i \left[\frac{1+e^2}{e^4}\right]. \end{aligned}$$

$$\oint_C \frac{e^{2z}\,dz}{(z+2)(z+1)^2} = 2\pi i \left(\frac{1+e^2}{e^4}\right)$$

Problem 2

Evaluate $\displaystyle\oint_C \frac{dz}{(z^2+4)^2}$, where C is $|z-i|=2$.

Solution:

Given

$$I = \oint_C \frac{dz}{(z^2+4)^2} = \oint_C f(z)\,dz, \qquad \text{where } f(z) = \frac{1}{(z^2+4)^2} \text{ and } C \text{ is } |z-i|=2.$$

By Cauchy's residue theorem,

$$I = 2\pi i \left(\text{Sum of the residues of } f(z) \text{ at its isolated singular points inside } C\right). \qquad (1)$$

Poles. Equating the denominator to zero:

$$\begin{aligned} (z^2+4)^2 = 0 \;&\Rightarrow\; \left(z^2-(2i)^2\right)^2 = 0 \qquad (\because\ (2i)^2 = -4) \\ &\Rightarrow\; \left((z+2i)(z-2i)\right)^2 = 0 \\ &\Rightarrow\; z = 2i,\ 2i,\ -2i,\ -2i. \end{aligned}$$

$\Rightarrow z = 2i$ and $z = -2i$ are poles of order 2.

Location. C is $|z-i|=2$ (the circle with centre i and radius 2).

When $z = 2i$:

$$|z-i| = |2i-i| = |i| = 1 < 2$$
$\Rightarrow z = 2i$ lies inside C.

When $z = -2i$:

$$|z-i| = |-2i-i| = |-3i| = 3 > 2$$
$\Rightarrow z = -2i$ lies outside C.

$\Rightarrow z = 2i$ is the only pole that lies inside C.

Residue at $z = 2i$ (pole of order 2).

$$\begin{aligned} \left[\operatorname{Res} f(z)\right]_{z=2i} &= \frac{1}{(2-1)!}\lim_{z\to 2i}\frac{d}{dz}\left[(z-2i)^2 f(z)\right] \\ &= \lim_{z\to 2i}\frac{d}{dz}\left[(z-2i)^2\frac{1}{(z^2+4)^2}\right] \\ &= \lim_{z\to 2i}\frac{d}{dz}\left[(z-2i)^2\frac{1}{(z+2i)^2(z-2i)^2}\right] \end{aligned}$$

$$= \lim_{z\to 2i} \frac{d}{dz}\left[\frac{1}{(z+2i)^2}\right] = \lim_{z\to 2i} \frac{d}{dz}(z+2i)^{-2}$$
$$= \lim_{z\to 2i} (-2)(z+2i)^{-3} = \lim_{z\to 2i} \frac{-2}{(z+2i)^3}$$
$$= \frac{-2}{(2i+2i)^3} = \frac{-2}{(4i)^3} = \frac{-2}{64\,i^3} = \frac{-2}{64(-i)} \qquad (\because\ i^3 = i^2\cdot i = -i)$$
$$= \frac{1}{32\,i}.$$

$$\Big[\operatorname{Res} f(z)\Big]_{z=2i} = \frac{1}{32\,i} \left(= -\frac{i}{32}\right)$$

Result. From (1),

$$I = 2\pi i \Big[\operatorname{Res} f(z)\Big]_{z=2i} = 2\pi i \left(\frac{-2}{64(-i)}\right) = 2\pi i \cdot \frac{1}{32\,i} = \frac{2\pi}{32} = \frac{\pi}{16}.$$

$$\oint_C \frac{dz}{(z^2+4)^2} = \frac{\pi}{16}$$

Problem 3

Evaluate $\displaystyle\oint_C \frac{z^2\,dz}{(z+2)^2(z^2+4)}$, where C is $|z| = 3$.

Solution:

Given

$$I = \oint_C \frac{z^2\,dz}{(z+2)^2(z^2+4)} = \oint_C f(z)\,dz, \qquad \text{where } f(z) = \frac{z^2}{(z+2)^2(z^2+4)} \text{ and } C \text{ is } |z| = 3.$$

Poles. Equating the denominator to zero:

$$(z+2)^2(z^2+4) = 0 \ \Rightarrow\ (z+2)^2(z+2i)(z-2i) = 0$$
$$\Rightarrow\ z = -2,\ -2,\ 2i,\ -2i.$$

$\Rightarrow z = -2$ is a pole of order 2; $z = 2i$ and $z = -2i$ are simple poles.

Location. C is $|z| = 3$.

$$z = -2: \quad |-2| = 2 < 3, \qquad z = 2i: \quad |2i| = 2 < 3, \qquad z = -2i: \quad |-2i| = 2 < 3.$$

$\Rightarrow$ All three poles $z = -2,\ 2i,\ -2i$ lie inside C. Hence, by Cauchy's residue theorem,

$$I = 2\pi i \left[\Big[\operatorname{Res} f(z)\Big]_{z=-2} + \Big[\operatorname{Res} f(z)\Big]_{z=2i} + \Big[\operatorname{Res} f(z)\Big]_{z=-2i}\right]. \tag{1}$$

Residue at $z = -2$ (pole of order 2).

$$\Big[\operatorname{Res} f(z)\Big]_{z=-2} = \frac{1}{1!}\lim_{z\to -2}\frac{d}{dz}\Big[(z-(-2))^2 f(z)\Big] = \lim_{z\to -2}\frac{d}{dz}\left[(z+2)^2\,\frac{z^2}{(z+2)^2(z^2+4)}\right]$$

$$= \lim_{z\to -2} \frac{d}{dz}\left[\frac{z^2}{z^2+4}\right] = \lim_{z\to -2}\left[\frac{(z^2+4)(2z)-z^2(2z)}{(z^2+4)^2}\right]$$
$$= \lim_{z\to -2}\left[\frac{2z^3+8z-2z^3}{(z^2+4)^2}\right] = \lim_{z\to -2}\frac{8z}{(z^2+4)^2}$$
$$= \frac{8(-2)}{\left((-2)^2+4\right)^2} = \frac{-16}{(4+4)^2} = \frac{-16}{64}.$$

$$\Big[\operatorname{Res} f(z)\Big]_{z=-2} = -\frac{1}{4} \tag{i}$$

Residue at $z = 2i$ (simple pole).

$$\Big[\operatorname{Res} f(z)\Big]_{z=2i} = \lim_{z\to 2i}(z-2i)f(z) = \lim_{z\to 2i}(z-2i)\,\frac{z^2}{(z+2)^2(z^2+4)}$$
$$= \lim_{z\to 2i}(z-2i)\,\frac{z^2}{(z+2)^2(z+2i)(z-2i)} = \lim_{z\to 2i}\frac{z^2}{(z+2)^2(z+2i)}$$
$$= \frac{(2i)^2}{(2i+2)^2(2i+2i)} = \frac{-4}{(2+2i)^2(4i)}.$$

Now $(2+2i)^2 = 4+8i+4i^2 = 4+8i-4 = 8i$. Therefore

$$\Big[\operatorname{Res} f(z)\Big]_{z=2i} = \frac{-4}{(8i)(4i)} = \frac{-4}{32\,i^2} = \frac{-4}{32(-1)} = \frac{4}{32}.$$

$$\Big[\operatorname{Res} f(z)\Big]_{z=2i} = \frac{1}{8} \tag{ii}$$

Residue at $z = -2i$ (simple pole).

$$\Big[\operatorname{Res} f(z)\Big]_{z=-2i} = \lim_{z\to -2i}\big(z-(-2i)\big)f(z) = \lim_{z\to -2i}(z+2i)\,\frac{z^2}{(z+2)^2(z^2+4)}$$
$$= \lim_{z\to -2i}(z+2i)\,\frac{z^2}{(z+2)^2(z+2i)(z-2i)} = \lim_{z\to -2i}\frac{z^2}{(z+2)^2(z-2i)}$$
$$= \frac{(-2i)^2}{(-2i+2)^2(-2i-2i)} = \frac{-4}{(2-2i)^2(-4i)}.$$

Now $(2-2i)^2 = 4-8i+4i^2 = 4-8i-4 = -8i$. Therefore

$$\Big[\operatorname{Res} f(z)\Big]_{z=-2i} = \frac{-4}{(-8i)(-4i)} = \frac{-4}{32\,i^2} = \frac{-4}{32(-1)} = \frac{4}{32}.$$

$$\Big[\operatorname{Res} f(z)\Big]_{z=-2i} = \frac{1}{8} \tag{iii}$$

Result. Substituting (i), (ii) and (iii) in (1):

$$I = 2\pi i\left[-\frac{1}{4}+\frac{1}{8}+\frac{1}{8}\right] = 2\pi i\left[\frac{-2+1+1}{8}\right] = 2\pi i\,(0) = 0.$$

$$\oint_C \frac{z^2\,dz}{(z+2)^2(z^2+4)} = 0$$

Problem 4

Evaluate $\displaystyle\oint_C \frac{\coth z}{z-i}\,dz$, where C is $|z|=2$.

Solution:

Given

$$I=\oint_C \frac{\coth z}{z-i}\,dz=\oint_C \frac{\cosh z}{\sinh z\,(z-i)}\,dz=\oint_C f(z)\,dz,$$

where $f(z)=\dfrac{\cosh z}{\sinh z\,(z-i)}$ and C is $|z|=2$. By Cauchy's residue theorem,

$$I=2\pi i\,\big(\text{Sum of the residues of } f(z) \text{ at its isolated singular points inside } C\big). \qquad (1)$$

Poles. The poles are given by $\sinh z\,(z-i)=0$

$$\Rightarrow \quad z-i=0 \quad \text{or} \quad \sinh z=0.$$

(a) $z-i=0$

$\Rightarrow z=i$ is a simple pole.

(b) $\sinh z=0$

$$\Rightarrow \frac{e^z-e^{-z}}{2}=0 \Rightarrow e^z=e^{-z} \Rightarrow e^z=\frac{1}{e^z}$$
$$\Rightarrow e^{2z}=1=\cos 2n\pi+i\sin 2n\pi, \quad n\in\mathbb{Z}$$
$$\Rightarrow e^{2z}=e^{i2n\pi} \Rightarrow 2z=i\,2n\pi \Rightarrow z=in\pi$$
$$\Rightarrow z=0,\ \pm i\pi,\ \pm 2i\pi,\ \pm 3i\pi,\ \ldots$$

Each zero of $\sinh z$ is simple, since $\dfrac{d}{dz}\sinh z=\cosh z$ and $\cosh(in\pi)=\cos n\pi=(-1)^n\neq 0$. So these are all simple poles.

Location. C is $|z|=2$.

- $z=i$: $|i|=1<2$ $\Rightarrow$ lies inside C.
- $z=0$: $|0|=0<2$ $\Rightarrow$ lies inside C.
- $z=\pm i\pi,\ \pm 2i\pi,\ldots$: $|z|\geq \pi\approx 3.14>2$ $\Rightarrow$ lie outside C.

$\Rightarrow$ The simple poles $z=i$ and $z=0$ lie inside C, so

$$I=2\pi i\left[\Big[\operatorname{Res} f(z)\Big]_{z=i}+\Big[\operatorname{Res} f(z)\Big]_{z=0}\right]. \qquad (1')$$

Residue at $z=i$ (simple pole).

$$\Big[\operatorname{Res} f(z)\Big]_{z=i}=\lim_{z\to i}(z-i)f(z)=\lim_{z\to i}(z-i)\frac{\coth z}{z-i}=\lim_{z\to i}\coth z=\coth i.$$

Since $\cosh i=\cos 1$ and $\sinh i=i\sin 1$, $\coth i=\dfrac{\cos 1}{i\sin 1}=-i\cot 1$.

$$\Big[\operatorname{Res} f(z)\Big]_{z=i}=\coth i \quad (=-i\cot 1) \qquad (i)$$

Residue at $z = 0$ (simple pole).

$$\Big[\operatorname{Res} f(z)\Big]_{z=0} = \lim_{z\to 0}(z-0)f(z) = \lim_{z\to 0} z\,\frac{\coth z}{z-i} = \lim_{z\to 0}\frac{z\cosh z}{\sinh z\,(z-i)} \qquad \left(\text{form } \frac{0}{0}\right)$$

Applying L'Hospital's rule (differentiate numerator and denominator):

$$\begin{aligned}\Big[\operatorname{Res} f(z)\Big]_{z=0} &= \lim_{z\to 0}\frac{\cosh z + z\sinh z}{\cosh z\,(z-i)+\sinh z\,(1)}\\ &= \frac{\cosh 0 + 0\cdot\sinh 0}{\cosh 0\,(0-i)+\sinh 0} = \frac{1+0}{(1)(-i)+0} = \frac{1}{-i} = -\frac{1}{i} = i \qquad \left(\because\ \frac{1}{i} = -i\right).\end{aligned}$$

$$\Big[\operatorname{Res} f(z)\Big]_{z=0} = i \tag{ii}$$

Result. Substituting (i) and (ii) in (1′):

$$I = 2\pi i\,(\coth i + i) = 2\pi i\,(-i\cot 1 + i) = 2\pi\cot 1 - 2\pi.$$

$$\oint_C \frac{\coth z}{z-i}\,dz = 2\pi i\,(\coth i + i) = 2\pi(\cot 1 - 1)$$

4.3 Exercises (with solutions)

Exercise 1

Evaluate $\oint_C \dfrac{dz}{\cosh z}$, where C is $|z| = 2$.

Solution:

Let $f(z) = \dfrac{1}{\cosh z}$ and C be $|z| = 2$.

Poles. $\cosh z = 0$

$$\Rightarrow \frac{e^z + e^{-z}}{2} = 0 \Rightarrow e^z = -e^{-z} \Rightarrow e^{2z} = -1 = \cos(2n+1)\pi + i\sin(2n+1)\pi = e^{i(2n+1)\pi}$$

$$\Rightarrow 2z = i(2n+1)\pi \Rightarrow z = \frac{(2n+1)\pi i}{2}, \quad n \in \mathbb{Z} \Rightarrow z = \pm\frac{\pi i}{2}, \pm\frac{3\pi i}{2}, \ldots$$

These are simple poles, since $\frac{d}{dz}\cosh z = \sinh z$ and $\sinh\left(\frac{(2n+1)\pi i}{2}\right) = i\sin\frac{(2n+1)\pi}{2} = \pm i \neq 0$.

Location. $\left|\pm\frac{\pi i}{2}\right| = \frac{\pi}{2} \approx 1.57 < 2$ (inside C); $\left|\pm\frac{3\pi i}{2}\right| = \frac{3\pi}{2} \approx 4.71 > 2$ (outside C), and so on. So only $z = \frac{\pi i}{2}$ and $z = -\frac{\pi i}{2}$ lie inside C.

Residues. Use formula (3) with $\phi(z) = 1$ and $\psi(z) = \cosh z$, so $\psi'(z) = \sinh z$:

$$\Big[\operatorname{Res} f(z)\Big]_{z=\pi i/2} = \frac{\phi(\pi i/2)}{\psi'(\pi i/2)} = \frac{1}{\sinh(\pi i/2)} = \frac{1}{i\sin(\pi/2)} = \frac{1}{i} = -i,$$

$$\Big[\operatorname{Res} f(z)\Big]_{z=-\pi i/2} = \frac{1}{\sinh(-\pi i/2)} = \frac{1}{-i\sin(\pi/2)} = \frac{1}{-i} = i.$$

Result.

$$I = 2\pi i\left[\Big[\operatorname{Res} f(z)\Big]_{z=\pi i/2} + \Big[\operatorname{Res} f(z)\Big]_{z=-\pi i/2}\right] = 2\pi i\,(-i+i) = 2\pi i\,(0) = 0.$$

$$\oint_C \frac{dz}{\cosh z} = 0$$

Exercise 2

Evaluate $\oint_C z^{-3}e^{z^2}\,dz$, where C is $|z| = 1$.

Solution:

Let $f(z) = z^{-3}e^{z^2} = \dfrac{e^{z^2}}{z^3}$ and C be $|z| = 1$.

Poles. $z^3 = 0 \Rightarrow z = 0, 0, 0$, i.e. $z = 0$ is a pole of order 3, and $|0| = 0 < 1$, so it lies inside C.

Residue at $z = 0$ (pole of order 3). Using formula (2) with $n = 3$:

$$\Big[\operatorname{Res} f(z)\Big]_{z=0} = \frac{1}{(3-1)!}\lim_{z\to 0}\frac{d^2}{dz^2}\Big[(z-0)^3 f(z)\Big] = \frac{1}{2!}\lim_{z\to 0}\frac{d^2}{dz^2}\left[z^3\,\frac{e^{z^2}}{z^3}\right] = \frac{1}{2}\lim_{z\to 0}\frac{d^2}{dz^2}\Big[e^{z^2}\Big].$$

Now

$$\frac{d}{dz}e^{z^2} = 2z\,e^{z^2}, \qquad \frac{d^2}{dz^2}e^{z^2} = \frac{d}{dz}\left(2z\,e^{z^2}\right) = 2e^{z^2} + 2z\cdot 2z\,e^{z^2} = (2+4z^2)\,e^{z^2}.$$

Hence

$$\Big[\operatorname{Res} f(z)\Big]_{z=0} = \frac{1}{2}\lim_{z\to 0}(2+4z^2)e^{z^2} = \frac{1}{2}\,(2+0)\,e^0 = \frac{1}{2}\cdot 2 = 1.$$

Check by Laurent series: $\dfrac{e^{z^2}}{z^3} = \dfrac{1}{z^3}\left[1 + z^2 + \dfrac{z^4}{2!} + \cdots\right] = \dfrac{1}{z^3} + \dfrac{1}{z} + \dfrac{z}{2} + \cdots$; the coefficient of $\dfrac{1}{z}$ is 1. ✓

Result.

$$I = 2\pi i\left[\operatorname{Res} f(z)\right]_{z=0} = 2\pi i\,(1) = 2\pi i.$$

$$\oint_C z^{-3}e^{z^2}\,dz = 2\pi i$$

Exercise 3

Evaluate $\displaystyle\oint_C z\,e^{1/z}\,dz$, where C is $|z| = 2$.

Solution:

Let $f(z) = z\,e^{1/z}$ and C be $|z| = 2$. The only singular point is $z = 0$, and $|0| < 2$, so it lies inside C.

Nature of $z = 0$. Expanding about $z = 0$:

$$f(z) = z\,e^{1/z} = z\left[1 + \frac{1}{z} + \frac{1}{2!\,z^2} + \frac{1}{3!\,z^3} + \frac{1}{4!\,z^4} + \cdots\right]$$
$$= z + 1 + \frac{1}{2!\,z} + \frac{1}{3!\,z^2} + \frac{1}{4!\,z^3} + \cdots$$

The principal part has infinitely many terms, so $z = 0$ is an **essential singularity**. The limit formulae cannot be used; the residue is the coefficient of $\dfrac{1}{z}$:

$$\Big[\operatorname{Res} f(z)\Big]_{z=0} = \frac{1}{2!} = \frac{1}{2}.$$

Result.

$$I = 2\pi i\left[\operatorname{Res} f(z)\right]_{z=0} = 2\pi i\cdot\frac{1}{2} = \pi i.$$

$$\oint_C z\,e^{1/z}\,dz = \pi i$$

5 Evaluation of Real Definite Integrals

Contour Integration – Type I & Type II

5.1 Evaluation of Real Definite Integrals

One of the important applications of the theory of residues consists in the evaluation of certain types of real definite integrals.

Contour integration

These integrals are evaluated by making the path of integration a suitable contour in the complex plane. This process is called contour integration.

5.2 Type I : $\int_0^{2\pi} f(\cos\theta, \sin\theta)\, d\theta$

Method

Type I

Integrals of the type $\displaystyle\int_0^{2\pi} f(\cos\theta, \sin\theta)\, d\theta$, where the integrand is a rational function of $\cos\theta$ and $\sin\theta$.

This type of integral can be evaluated by putting $z = e^{i\theta}$. Then

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$
$$\Rightarrow \cos\theta = \frac{z + \frac{1}{z}}{2}$$
$$\Rightarrow \cos\theta = \frac{z^2 + 1}{2z}$$

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \Rightarrow \sin\theta = \frac{z - \frac{1}{z}}{2i} \Rightarrow \sin\theta = \frac{z^2 - 1}{2iz}$$

Further,

$$z = e^{i\theta} \Rightarrow dz = e^{i\theta}\, i\, d\theta \Rightarrow d\theta = \frac{dz}{iz}$$

Moreover, $z = e^{i\theta} \Rightarrow |z| = 1 \Rightarrow C$ is the unit circle $|z| = 1$.

Summary of substitutions ($z = e^{i\theta}$)

$$\cos\theta = \frac{z^2+1}{2z}, \qquad \sin\theta = \frac{z^2-1}{2iz}, \qquad d\theta = \frac{dz}{iz}, \qquad C: \ |z| = 1$$

Result

Hence

$$\int_0^{2\pi} f(\cos\theta, \sin\theta)\, d\theta = \int_C f(z)\, dz, \quad \text{where } C \text{ is } |z| = 1$$
$$= 2\pi i \left(\text{Sum of the residues of } f(z)\right)$$

(by Cauchy's residue theorem, the residues being those at the poles inside C).

Problem 1 : $\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta} d\theta$

Problem 1

Evaluate $\displaystyle\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta}\, d\theta.$

Solution. Given

$$I = \int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta}\, d\theta \tag{1}$$

Put $z = e^{i\theta} \Rightarrow \cos\theta = \dfrac{z^2+1}{2z}$ and $\sin\theta = \dfrac{z^2-1}{2iz}$.

Further,

$$\cos 2\theta = \frac{e^{i2\theta} + e^{-i2\theta}}{2} = \frac{z^2 + z^{-2}}{2}$$
$$\cos 2\theta = \frac{z^2 + \frac{1}{z^2}}{2} = \frac{z^4+1}{2z^2} \quad \text{and} \quad d\theta = \frac{dz}{iz}$$

and $C: |z| = 1$, a unit circle.

$\therefore$ (1) $\Rightarrow$

$$I = \int_C \frac{\frac{z^4+1}{2z^2}}{5+4\left(\frac{z^2+1}{2z}\right)} \frac{dz}{iz}$$
$$= \int_C \frac{\frac{z^4+1}{2z^2}}{\frac{5z+2z^2+2}{z}} \frac{dz}{iz}$$

$$I = \int_C \frac{z^4+1}{2z^2} \times \frac{z}{2z^2+5z+2} \times \frac{dz}{iz}$$
$$= \frac{1}{2i} \int_C \frac{z^4+1}{z^2(2z^2+5z+2)}\, dz$$
$$= \frac{1}{2i} \int_C f(z)\, dz, \quad \text{where } f(z) = \frac{z^4+1}{z^2(2z^2+5z+2)}$$

$$I = \frac{1}{2i}\Big(2\pi i \,\big(\text{Sum of the residues of } f(z) \text{ at its poles}\big)\Big)$$
$$I = \pi \,\big[\text{Sum of the residues of } f(z) \text{ at its poles}\big] \tag{2}$$

Now $f(z) = \dfrac{z^4+1}{z^2(2z^2+5z+2)}$. The poles are given by $z^2(2z^2+5z+2)=0$

$$\Rightarrow \; z^2 = 0 \quad \text{or} \quad 2z^2+5z+2=0$$
$$\Rightarrow \; z = 0, 0 \quad \text{or} \quad 2z^2+4z+z+2=0$$
$$2z(z+2)+(z+2)=0$$
$$(2z+1)(z+2)=0$$
$$\boxed{z = -\tfrac{1}{2},\, -2}$$

Since $C:\ |z|=1$,

$\Rightarrow$ $z=0$ and $z=-\frac{1}{2}$ are the only poles lying inside C ($|-2|=2>1$, so $z=-2$ lies outside C),

where $z=0$ is a pole of order 2 and $z=-\frac{1}{2}$ is a simple pole.

Let us find the residues at these two poles.

$$\Big(\operatorname{Res} f(z)\Big)_{z=0} = \frac{1}{1!}\lim_{z\to 0}\frac{d}{dz}\Big((z-0)^2 f(z)\Big) \qquad \text{(pole of order 2)}$$

$$\Big(\operatorname{Res} f(z)\Big)_{z=0} = \lim_{z\to 0}\frac{d}{dz}\left[z^2\,\frac{z^4+1}{z^2(2z^2+5z+2)}\right]$$
$$= \lim_{z\to 0}\frac{d}{dz}\left[\frac{z^4+1}{2z^2+5z+2}\right]$$

$$\Big(\operatorname{Res} f(z)\Big)_{z=0} = \lim_{z\to 0}\left(\frac{(2z^2+5z+2)(4z^3)-(z^4+1)(4z+5)}{(2z^2+5z+2)^2}\right)$$
$$= \frac{(2)(0)-(1)(5)}{(2)^2}$$
$$\Big(\operatorname{Res} f(z)\Big)_{z=0} = -\frac{5}{4} \tag{i}$$

Now, $z=-\frac{1}{2}$ is a simple pole:

$$\Big(\operatorname{Res} f(z)\Big)_{z=-\frac{1}{2}} = \lim_{z\to -\frac{1}{2}}\Big(z-\big(-\tfrac{1}{2}\big)\Big)f(z)$$
$$= \lim_{z\to -\frac{1}{2}}\Big(z+\tfrac{1}{2}\Big)\frac{z^4+1}{z^2(2z^2+5z+2)}$$

$$= \lim_{z\to -\frac{1}{2}} \left(z+\tfrac{1}{2}\right) \frac{z^4+1}{z^2\cdot 2\left(z+\frac{1}{2}\right)(z+2)}$$

(since $2z^2+5z+2 = (2z+1)(z+2) = 2\left(z+\frac{1}{2}\right)(z+2)$)

$$\left(\operatorname{Res} f(z)\right)_{z=-\frac{1}{2}} = \lim_{z\to -\frac{1}{2}} \frac{z^4+1}{2z^2(z+2)} = \frac{\left(-\frac{1}{2}\right)^4+1}{\left(-\frac{1}{2}\right)^2\left(2\left(-\frac{1}{2}+2\right)\right)}$$

$$= \frac{\frac{1}{16}+1}{\frac{1}{4}\cdot 2\cdot \frac{3}{2}} = \frac{17/16}{3/4}$$

$$\Rightarrow \left(\operatorname{Res} f(z)\right)_{z=-\frac{1}{2}} = \frac{17}{16}\times\frac{4}{3} = \frac{17}{12} \qquad \text{(ii)}$$

$\therefore$ Substituting (i) and (ii) in (2), we get

$$I = \pi\left[\left(\operatorname{Res} f(z)\right)_{z=0} + \left(\operatorname{Res} f(z)\right)_{z=-\frac{1}{2}}\right]$$

$$= \pi\left[-\frac{5}{4}+\frac{17}{12}\right] = \pi\left[\frac{-15+17}{12}\right] = \pi\cdot\frac{2}{12}$$

$$\boxed{I = \frac{\pi}{6}}$$

Problem 2 : $\int_0^{\pi} \frac{d\theta}{3+2\cos\theta}$

Problem 2

Evaluate $\displaystyle\int_0^{\pi} \frac{d\theta}{3+2\cos\theta}$.

Solution. Given

$$I = \int_0^{\pi} \frac{d\theta}{3+2\cos\theta} = \frac{1}{2}\int_0^{2\pi} \frac{d\theta}{3+2\cos\theta} \qquad (1)$$

$\left[\because\ f(2a-\theta) = f(\theta)\ \Rightarrow\ \displaystyle\int_0^{2a} f(\theta)\,d\theta = 2\int_0^{a} f(\theta)\,d\theta\right]$

Put $z = e^{i\theta}$, we get $\cos\theta = \dfrac{z^2+1}{2z}$, $d\theta = \dfrac{dz}{iz}$ and C is $|z| = 1$.

$\therefore$ (1) $\Rightarrow$

$$I = \frac{1}{2}\int_C \frac{dz/iz}{3+2\left(\frac{z^2+1}{2z}\right)} = \frac{1}{2i}\int_C \frac{dz/z}{\frac{3z+z^2+1}{z}}$$

$$I = \frac{1}{2i}\int_C \frac{dz}{z}\times\frac{z}{z^2+3z+1}$$

$$= \frac{1}{2i}\int_C \frac{dz}{z^2+3z+1}$$

$$= \frac{1}{2i}\int_C f(z)\,dz, \quad \text{where } f(z) = \frac{1}{z^2+3z+1}$$

$$I = \frac{1}{2i}\Big(2\pi i\,(\text{Sum of the residues of } f(z)$$

$$\text{at its isolated singular points)}\Big)$$

$$I = \pi\,[\text{Sum of the residues of } f(z) \text{ at its isolated singular points}] \tag{2}$$

Now $f(z) = \dfrac{1}{z^2+3z+1}$. The poles are given by $z^2+3z+1=0$:

$$z = \frac{-3 \pm \sqrt{9-4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$$

$\Rightarrow\ z = \dfrac{-3+\sqrt{5}}{2},\ \dfrac{-3-\sqrt{5}}{2}$ are simple poles.

But C is $|z|=1 \Rightarrow z = \dfrac{-3+\sqrt{5}}{2}$ is the only pole lying inside C.

$\left(\left|\frac{-3+\sqrt{5}}{2}\right| \approx 0.382 < 1, \quad \left|\frac{-3-\sqrt{5}}{2}\right| \approx 2.618 > 1\right)$

$$\therefore\ \Big(\text{Res}\, f(z)\Big)_{z=\frac{-3+\sqrt{5}}{2}} = \lim_{z\to\frac{-3+\sqrt{5}}{2}} \left(z - \tfrac{-3+\sqrt{5}}{2}\right) f(z)$$

$$= \lim_{z\to\alpha} (z-\alpha)\frac{1}{z^2+3z+1}$$

where $\alpha = \dfrac{-3+\sqrt{5}}{2}$.

$$= \lim_{z\to\alpha} (z-\alpha)\frac{1}{(z-\alpha)(z-\beta)} \qquad \left(\beta = \tfrac{-3-\sqrt{5}}{2}\right)$$

$$= \lim_{z\to\frac{-3+\sqrt{5}}{2}} \frac{1}{z - \frac{-3-\sqrt{5}}{2}}$$

$$\Big(\text{Res}\, f(z)\Big)_{z=\frac{-3+\sqrt{5}}{2}} = \frac{1}{\frac{-3+\sqrt{5}}{2} + \frac{3+\sqrt{5}}{2}} = \frac{1}{\frac{2\sqrt{5}}{2}} = \frac{1}{\sqrt{5}} \tag{i}$$

Substituting (i) in (2), we get

$$\Rightarrow\ I = \pi\left[\Big(\text{Res}\, f(z)\Big)_{z=\frac{-3+\sqrt{5}}{2}}\right]$$

$$= \pi\left[\frac{1}{\sqrt{5}}\right]$$

$$\Rightarrow\ \boxed{I = \frac{\pi}{\sqrt{5}}}$$

Exercise 3 : $\int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta$

Ex. 3

Evaluate $\displaystyle\int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta.$

Note

(Set as an exercise in the notes; the solution below is worked out for completeness.)

Solution. Given

$$I = \int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta = \frac{1}{2}\int_0^{2\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta \tag{1}$$

$\left[\because\ f(2\pi-\theta) = f(\theta)\right]$

Put $z = e^{i\theta}$: $\cos\theta = \dfrac{z^2+1}{2z}$, $d\theta = \dfrac{dz}{iz}$, $C : |z| = 1$.

$$1+2\cos\theta = 1+2\left(\frac{z^2+1}{2z}\right) = 1+\frac{z^2+1}{z} = \frac{z^2+z+1}{z}$$

$$5+4\cos\theta = 5+4\left(\frac{z^2+1}{2z}\right) = 5+\frac{2(z^2+1)}{z} = \frac{2z^2+5z+2}{z}$$

$\therefore$ (1) $\Rightarrow$

$$I = \frac{1}{2}\int_C \frac{\dfrac{z^2+z+1}{z}}{\dfrac{2z^2+5z+2}{z}}\,\frac{dz}{iz} = \frac{1}{2}\int_C \frac{z^2+z+1}{z}\times\frac{z}{2z^2+5z+2}\times\frac{dz}{iz}$$

$$= \frac{1}{2i}\int_C \frac{z^2+z+1}{z(2z^2+5z+2)}\,dz = \frac{1}{2i}\int_C f(z)\,dz$$

where $f(z) = \dfrac{z^2+z+1}{z(2z^2+5z+2)}$.

$$I = \frac{1}{2i}\Big(2\pi i\,\big(\text{Sum of the residues of } f(z) \text{ at its poles}\big)\Big)$$

$$I = \pi\left[\text{Sum of the residues of } f(z) \text{ at its poles}\right] \tag{2}$$

Poles: $z(2z^2+5z+2) = 0 \Rightarrow z(2z+1)(z+2) = 0$

$$\Rightarrow\ z = 0,\ -\tfrac{1}{2},\ -2$$

Since $C : |z| = 1$, $z = 0$ and $z = -\frac{1}{2}$ (both simple poles) lie inside C; $z = -2$ lies outside C.

$$\Big(\operatorname{Res} f(z)\Big)_{z=0} = \lim_{z\to 0} z\,f(z) = \lim_{z\to 0}\frac{z^2+z+1}{2z^2+5z+2} = \frac{1}{2} \tag{i}$$

$$\Big(\operatorname{Res} f(z)\Big)_{z=-\frac{1}{2}} = \lim_{z\to -\frac{1}{2}}\Big(z+\tfrac{1}{2}\Big)\frac{z^2+z+1}{z\cdot 2\big(z+\frac{1}{2}\big)(z+2)}$$

$$= \lim_{z\to -\frac{1}{2}}\frac{z^2+z+1}{2z(z+2)}$$

$$= \frac{\frac{1}{4}-\frac{1}{2}+1}{2\big(-\frac{1}{2}\big)\big(\frac{3}{2}\big)} = \frac{3/4}{-3/2} = -\frac{1}{2} \tag{ii}$$

Substituting (i) and (ii) in (2), we get

$$I = \pi\left[\Big(\operatorname{Res} f(z)\Big)_{z=0} + \Big(\operatorname{Res} f(z)\Big)_{z=-\frac{1}{2}}\right]$$

$$= \pi\left[\frac{1}{2}-\frac{1}{2}\right]$$

$$\boxed{I = 0}$$

Recall : Type I method

Recall : Type I method

Integrals of the form $\int_0^{2\pi} f(\cos\theta, \sin\theta)\, d\theta$ can be converted into $\int_C f(z)\, dz$ by putting $z = e^{i\theta}$

$$\Rightarrow \cos\theta = \frac{z^2+1}{2z}, \qquad \sin\theta = \frac{z^2-1}{2iz}, \qquad d\theta = \frac{dz}{iz}$$

and C is $|z| = 1$.

Further, we can apply Cauchy's residue theorem.

Problem : $\int_0^{\pi} \frac{a\, d\phi}{a^2+\sin^2\phi}$

Problem

Show that $\int_0^{\pi} \frac{a\, d\phi}{a^2+\sin^2\phi} = \frac{\pi}{\sqrt{1+a^2}}$, where $a > 0$.

Solution. Given

$$I = \int_0^{\pi} \frac{a\, d\phi}{a^2+\sin^2\phi} = \int_0^{\pi} \frac{a\, d\phi}{a^2 + \left(\frac{1-\cos 2\phi}{2}\right)}$$

$$I = \int_0^{\pi} \frac{2a\, d\phi}{2a^2+1-\cos 2\phi}$$

Let $2\phi = \theta \Rightarrow 2\, d\phi = d\theta$. When $\phi = 0 \Rightarrow \theta = 0$; $\phi = \pi \Rightarrow \theta = 2\pi$.

$$I = \int_0^{2\pi} \frac{a\, d\theta}{2a^2+1-\cos\theta} \tag{1}$$

Put $z = e^{i\theta} \Rightarrow \cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2} = \frac{z+\frac{1}{z}}{2} = \frac{z^2+1}{2z}$

$dz = e^{i\theta}\, i\, d\theta \Rightarrow d\theta = \frac{dz}{iz}$ and $C : |z| = 1$.

$\therefore$ (1) $\Rightarrow$

$$I = \int_C \frac{a\, d\theta}{2a^2+1-\left(\frac{z^2+1}{2z}\right)} = \int_C \frac{a\, dz/iz}{\left(4a^2z+2z-z^2-1\right)/2z}$$

$$= \frac{2a}{i}\int_C \frac{dz}{-z^2+(4a^2+2)z-1}$$

$$= \frac{2a}{-i}\int_C \frac{dz}{z^2-2(1+2a^2)z+1}$$

$$I = 2ia\int_C f(z)\, dz \qquad \left(\because \tfrac{1}{-i} = i\right)$$

$$= 2ia\left[2\pi i\,(\text{Sum of the residues of } f(z) \text{ at its poles})\right]$$

$$I = -4a\pi\,[\text{Sum of the residues of } f(z) \text{ at its poles}] \tag{2}$$

where $f(z) = \frac{1}{z^2-2(1+2a^2)z+1}$ and C is $|z| = 1$.

Now $f(z) = \dfrac{1}{z^2 - 2(1+2a^2)z + 1}$. Poles are given by $z^2 - 2(1+2a^2)z + 1 = 0$:

$$z = \frac{2(1+2a^2) \pm \sqrt{4(1+2a^2)^2 - 4}}{2}$$
$$= \frac{2(1+2a^2) \pm 2\sqrt{1+4a^4+4a^2-1}}{2}$$
$$z = 1 + 2a^2 \pm 2a\sqrt{a^2+1}$$

$\Rightarrow$ We have two roots as well as poles of $f(z)$:

$$\alpha = 2a^2 + 1 + 2a\sqrt{a^2+1} \quad \text{and} \quad \beta = 2a^2 + 1 - 2a\sqrt{a^2+1}$$

Given $a > 0$. When we take $a = 1$, we get

$$\alpha = 3 + 2\sqrt{2} \quad \text{and} \quad \beta = 3 - 2\sqrt{2} \;\Rightarrow\; |\alpha| > 1 \text{ and } |\beta| < 1$$

(In general $\alpha\beta = 1$ and $\alpha > 1$ for $a > 0$, so $0 < \beta < 1$.)

$\Rightarrow$ $\beta = 2a^2 + 1 - 2a\sqrt{a^2+1}$ lies inside C.

Let us find Res $f(z)$ at $z = \beta$.

Now, $z = \beta$ is a simple pole:

$$\Big(\text{Res } f(z)\Big)_{z=\beta} = \lim_{z\to\beta}(z-\beta)f(z)$$
$$= \lim_{z\to\beta}(z-\beta)\frac{1}{z^2 - 2(2a^2+1)z + 1}$$
$$= \lim_{z\to\beta}(z-\beta)\frac{1}{(z-\alpha)(z-\beta)} \quad (\because\ \alpha, \beta \text{ are roots})$$
$$= \frac{1}{\beta - \alpha}$$

$$\Big(\text{Res } f(z)\Big)_{z=\beta} = \frac{1}{\left(2a^2+1-2a\sqrt{a^2+1}\right) - \left(2a^2+1+2a\sqrt{a^2+1}\right)}$$
$(\because$ substituting β and $\alpha)$
$$\Big(\text{Res } f(z)\Big)_{z=\beta} = \frac{1}{-4a\sqrt{a^2+1}}$$

$\therefore$ (2) $\Rightarrow$

$$I = -4a\pi \Big(\text{Res } f(z)\Big)_{z=\beta}$$
$$= -4a\pi \left(\frac{-1}{4a\sqrt{a^2+1}}\right)$$
$$\boxed{I = \frac{\pi}{\sqrt{1+a^2}}}$$

Exercises (Type I)

Exercises (Type I)

1. Show that $\displaystyle\int_0^{2\pi} \frac{d\theta}{a + b\cos\theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}, \quad a > b > 0.$

2. Show that $\displaystyle\int_0^{2\pi} \frac{a\,d\theta}{a^2+\sin^2\theta} = \frac{2\pi}{\sqrt{1+a^2}}, \quad a>0.$

3. Show that $\displaystyle\int_0^{2\pi} \frac{\sin^2\theta\,d\theta}{a+b\cos\theta} = \frac{2\pi}{b^2}\left\{a-\sqrt{a^2-b^2}\right\}, \quad a>b>0.$

5.3 Type II : $\int_{-\infty}^{\infty} \frac{p(x)}{q(x)}\,dx$

Method

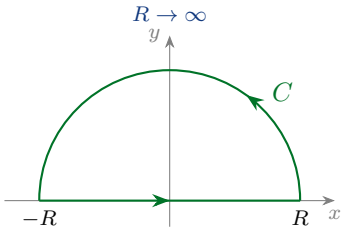
Type II

Integrals of the type $\displaystyle\int_{-\infty}^{\infty} \frac{p(x)}{q(x)}\,dx$, where deg $(q(x))$ is greater than deg $(p(x))$ by at least two, and $q(x)$ has no real roots.

This type of integral can be evaluated as follows:

$$\int_{-\infty}^{\infty} \frac{p(x)}{q(x)}\,dx = \int_C \frac{p(z)}{q(z)}\,dz$$

where C is the very large semicircle in the upper half plane.



$$\int_{-\infty}^{\infty} \frac{p(x)}{q(x)}\,dx = \int_C f(z)\,dz$$
$$= 2\pi i \left[\text{Sum of the residues of } f(z) \text{ at its poles}\right]$$

where $f(z) = \dfrac{p(z)}{q(z)}$
(poles lying inside C, i.e. in the upper half plane).

Problem 1 : $\int_{-\infty}^{\infty} \frac{x^2}{(x^2+a^2)(x^2+b^2)}dx$

Problem 1

Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{x^2}{(x^2+a^2)(x^2+b^2)}\,dx = \frac{\pi}{a+b}, \quad a,b>0.$

Solution. Given

$$I = \int_{-\infty}^{\infty} \frac{x^2}{(x^2+a^2)(x^2+b^2)}\,dx, \quad a,b>0$$

$$I = \int_C \frac{z^2\,dz}{(z^2+a^2)(z^2+b^2)} \qquad (\because \text{ by contour integration})$$

where C is the very large semicircle in the upper half plane.

$$= \int_C f(z)\,dz, \quad \text{where } f(z) = \frac{z^2}{(z^2+a^2)(z^2+b^2)}$$

$$I = 2\pi i \left[\text{Sum of the residues of } f(z) \text{ at its poles}\right] \tag{1}$$

Now $f(z) = \dfrac{z^2}{(z^2+a^2)(z^2+b^2)}$. Poles are given by $(z^2+a^2)(z^2+b^2) = 0$

$$(z^2+a^2) = 0 \quad \text{or} \quad z^2+b^2 = 0$$
$$(z+ai)(z-ai) = 0 \quad \text{or} \quad (z+bi)(z-bi) = 0$$
$$\Rightarrow z = ai,\ -ai \quad \text{or} \quad z = bi,\ -bi$$

$\Rightarrow$ $z = ai$ and $z = bi$ are poles of order 1 lying in C
($\because$ $a > 0$, $b > 0$ and C is the very large semicircle in the upper half plane).
$z = ai$ is a simple pole:

$$\therefore \left(\text{Res } f(z)\right)_{z=ai} = \lim_{z\to ai}(z-ai)f(z)$$
$$= \lim_{z\to ai}(z-ai)\frac{z^2}{(z^2+a^2)(z^2+b^2)}$$
$$= \lim_{z\to ai}(z-ai)\frac{z^2}{(z+ai)(z-ai)(z^2+b^2)}$$

$$\left(\text{Res } f(z)\right)_{z=ai} = \lim_{z\to ai}\frac{z^2}{(z+ai)(z^2+b^2)} = \frac{(ai)^2}{(2ai)\left((ai)^2+b^2\right)}$$
$$= \frac{-a^2}{2ai\,(b^2-a^2)}$$
$$\left(\text{Res } f(z)\right)_{z=ai} = \frac{a}{2i\,(a^2-b^2)} \tag{i}$$

Illy, $z = bi$ is a simple pole:

$$\left(\text{Res } f(z)\right)_{z=bi} = \lim_{z\to bi}(z-bi)f(z)$$
$$= \lim_{z\to bi}(z-bi)\frac{z^2}{(z^2+a^2)(z^2+b^2)}$$
$$= \lim_{z\to bi}(z-bi)\frac{z^2}{(z^2+a^2)(z+bi)(z-bi)}$$
$$= \frac{(bi)^2}{\left((bi)^2+a^2\right)(2bi)}$$

$$\left(\text{Res } f(z)\right)_{z=bi} = \frac{(bi)^2}{\left((bi)^2+a^2\right)(2bi)}$$
$$= \frac{-b^2}{(-b^2+a^2)(2bi)}$$
$$\left(\text{Res } f(z)\right)_{z=bi} = \frac{-b}{2i\,(a^2-b^2)} \tag{ii}$$

Substituting (i) and (ii) in (1), we get

$$I = 2\pi i\left[\left(\text{Res } f(z)\right)_{z=ai} + \left(\text{Res } f(z)\right)_{z=bi}\right]$$
$$= 2\pi i\left[\frac{a}{2i(a^2-b^2)} - \frac{b}{2i(a^2-b^2)}\right]$$

$$= \frac{\pi}{a^2 - b^2}\left[a - b\right] = \frac{\pi}{(a+b)(a-b)}(a-b)$$

$$\Rightarrow \boxed{I = \frac{\pi}{a+b}}, \quad a, b > 0$$

Problem 2 : $\int_0^\infty \frac{dx}{(x^2+a^2)^3}$

Problem 2

Evaluate $\displaystyle\int_0^\infty \frac{dx}{(x^2+a^2)^3}$, where $a > 0$.

Solution. Given

$$I = \int_0^\infty \frac{dx}{(x^2+a^2)^3}$$
$$= \frac{1}{2}\int_{-\infty}^\infty \frac{dx}{(x^2+a^2)^3}$$

$\left[\because \text{by even function } f(x), \int_{-\infty}^\infty f(x)\,dx = 2\int_0^\infty f(x)\,dx\right]$

$$I = \frac{1}{2}\int_C \frac{dz}{(z^2+a^2)^3}$$

where C is the very large semicircle in the upper half plane.

$$= \frac{1}{2}\int_C f(z)\,dz, \quad \text{where } f(z) = \frac{1}{(z^2+a^2)^3}$$
$$= \frac{1}{2}\Big[2\pi i\,(\text{Sum of the residues of } f(z) \text{ at its poles})\Big]$$
$$I = \pi i\left[\text{Sum of the residues of } f(z) \text{ at its poles}\right] \tag{1}$$

Now $f(z) = \dfrac{1}{(z^2+a^2)^3}$ $\Rightarrow$ poles are given by $(z^2+a^2)^3 = 0$

$$\Rightarrow \left[(z+ai)(z-ai)\right]^3 = 0$$
$$\Rightarrow z = -ai, -ai, -ai, ai, ai, ai$$

Since $a > 0$, $z = ai$, a pole of order 3, lies inside C ($\because$ upper half plane).

$$\therefore \left(\text{Res}\, f(z)\right)_{z=ai} = \frac{1}{2!}\lim_{z\to ai}\frac{d^2}{dz^2}\Big[(z-ai)^3 f(z)\Big] \quad \text{(pole of order 3)}$$

$$\left(\text{Res}\, f(z)\right)_{z=ai} = \frac{1}{2}\lim_{z\to ai}\frac{d^2}{dz^2}\left[(z-ai)^3\frac{1}{(z^2+a^2)^3}\right]$$
$$= \frac{1}{2}\lim_{z\to ai}\left(\frac{d}{dz}\left[\frac{d}{dz}\left[(z-ai)^3\frac{1}{(z+ai)^3(z-ai)^3}\right]\right]\right)$$
$$= \frac{1}{2}\lim_{z\to ai}\frac{d}{dz}\Big\{-3(z+ai)^{-4}(1)\Big\}$$

$$\left(\text{Res}\, f(z)\right)_{z=ai} = \frac{1}{2}\lim_{z\to ai}\Big[12(z+ai)^{-5}(1)\Big]$$

$$= \frac{1}{2}\lim_{z\to ai}\frac{12}{(z+ai)^5} = \frac{1}{2}\cdot\frac{12}{(2ai)^5}$$

$$= \frac{6}{32a^5 i} \qquad (\because\ i^5 = i^4\cdot i = i)$$

$$\left(\text{Res}\, f(z)\right)_{z=ai} = \frac{3}{16a^5 i}$$

$\therefore$ (1) $\Rightarrow$

$$I = \pi i\left[\left(\text{Res}\, f(z)\right)_{z=ai}\right]$$

$$= \pi i\left(\frac{3}{16a^5 i}\right)$$

$$\boxed{I = \frac{3\pi}{16a^5}}, \quad a > 0$$

Problem 3 : $\int_{-\infty}^{\infty}\frac{x^2dx}{(x^2+1)(x^2+2x+2)}$

Problem 3

Evaluate $\displaystyle\int_{-\infty}^{\infty}\frac{x^2\,dx}{(x^2+1)(x^2+2x+2)}$. **Ans:** $I = \frac{3\pi}{5}$

Note

(Only the answer is given in the notes; the solution below is worked out for completeness.)

Solution.

$$I = \int_C f(z)\,dz, \quad f(z) = \frac{z^2}{(z^2+1)(z^2+2z+2)}$$

where C is the very large semicircle in the upper half plane.

$$I = 2\pi i\left[\text{Sum of the residues of } f(z) \text{ at its poles}\right] \tag{1}$$

Poles are given by $(z^2+1)(z^2+2z+2) = 0$:

$$z^2+1 = 0 \ \Rightarrow\ z = \pm i$$

$$z^2+2z+2 = 0 \ \Rightarrow\ z = \frac{-2\pm\sqrt{4-8}}{2} = \frac{-2\pm 2i}{2} = -1\pm i$$

$\Rightarrow$ $z = i$ and $z = -1+i$ are simple poles lying inside C (upper half plane); $z = -i$, $z = -1-i$ lie outside C.

$$\left(\text{Res}\, f(z)\right)_{z=i} = \lim_{z\to i}(z-i)\frac{z^2}{(z+i)(z-i)(z^2+2z+2)}$$

$$= \frac{i^2}{(2i)(i^2+2i+2)}$$

$$= \frac{-1}{2i\,(1+2i)} = \frac{-1}{2i-4}$$

$$= \frac{1}{4-2i} = \frac{4+2i}{(4-2i)(4+2i)} = \frac{4+2i}{20}$$

$$\left(\text{Res}\, f(z)\right)_{z=i} = \frac{2+i}{10} \tag{i}$$

Since $z^2+2z+2=\big(z-(-1+i)\big)\big(z-(-1-i)\big)$,

$$\Big(\operatorname{Res} f(z)\Big)_{z=-1+i}=\lim_{z\to -1+i}\frac{z^2}{(z^2+1)\big(z-(-1-i)\big)}$$

At $z=-1+i$: $z^2=(-1+i)^2=1-2i+i^2=-2i$, $z^2+1=1-2i$, $z-(-1-i)=2i$.

$$\Big(\operatorname{Res} f(z)\Big)_{z=-1+i}=\frac{-2i}{(1-2i)(2i)}=\frac{-1}{1-2i}=\frac{-(1+2i)}{(1-2i)(1+2i)}$$

$$\Big(\operatorname{Res} f(z)\Big)_{z=-1+i}=\frac{-(1+2i)}{5}=\frac{-2-4i}{10} \qquad \text{(ii)}$$

Substituting (i) and (ii) in (1), we get

$$I=2\pi i\left[\frac{2+i}{10}+\frac{-2-4i}{10}\right]$$
$$=2\pi i\left[\frac{-3i}{10}\right]=\frac{-6\pi i^2}{10}=\frac{6\pi}{10}$$

$$\boxed{I=\frac{3\pi}{5}}$$

Problem : $\int_0^\infty \frac{x^2}{x^4+a^4}dx$

Problem

Show that $\displaystyle\int_0^\infty \frac{x^2\,dx}{x^4+a^4}=\frac{\pi}{2\sqrt{2}\,a}$, where $a>0$.

Solution. Given

$$I=\int_0^\infty \frac{x^2}{x^4+a^4}\,dx, \quad a>0$$
$$=\frac{1}{2}\int_{-\infty}^{\infty}\frac{x^2}{x^4+a^4}\,dx \qquad (\because \text{ even function})$$

$$I=\frac{1}{2}\int_C \frac{z^2}{z^4+a^4}\,dz$$

where C is the very large semicircle in the upper half plane.

$$=\frac{1}{2}\Big\{2\pi i\,(\text{Sum of the residues of } f(z) \text{ at its poles})\Big\}$$
$$I=\pi i\,[\text{Sum of the residues of } f(z) \text{ at its poles}] \qquad (1)$$

Now $f(z)=\dfrac{z^2}{z^4+a^4}$. Poles are given by $z^4+a^4=0$

$$\Rightarrow\ z^4=-a^4=a^4(-1)$$
$$=a^4(\cos\pi+i\sin\pi)$$

Remark

If we split $z^4+a^4=(z^2)^2+(a^2)^2=(z^2+a^2i)(z^2-a^2i)$, then $z^2=-a^2i\Rightarrow z=ia\sqrt{i}$ – *not suitable.*

But we have to get 4 roots, so we may consider $\cos 3\pi$, $\cos 5\pi$, $\cos 7\pi$; hence the above expression can be written as

$$\begin{aligned} z^4 &= a^4\big(\cos(2n+1)\pi + i\sin(2n+1)\pi\big), \quad n = 0,1,2,3 \\ &= a^4 e^{i(2n+1)\pi} \qquad (\because\ e^{i\theta} = \cos\theta + i\sin\theta) \\ \Rightarrow\ z &= a\, e^{i(2n+1)\pi/4}, \quad n = 0,1,2,3 \end{aligned}$$

$\Rightarrow\ z = ae^{i\pi/4},\ ae^{i3\pi/4},\ ae^{i5\pi/4},\ ae^{i7\pi/4}$ are the roots as well as poles.

① $z = ae^{i\pi/4} = a\Big(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\Big)$

$= a\Big(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\Big)$ [lies inside C]

② $z = ae^{i3\pi/4} = a\Big(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\Big)$

$= a\Big(\cos\big(\pi - \frac{\pi}{4}\big) + i\sin\big(\pi - \frac{\pi}{4}\big)\Big)$

$= a\Big(-\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\Big) = a\Big(-\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\Big)$ [lies inside C]

③ $z = ae^{i5\pi/4} = a\Big(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\Big)$

$= a\Big(\cos\big(\pi + \frac{\pi}{4}\big) + i\sin\big(\pi + \frac{\pi}{4}\big)\Big)$

$= a\Big(-\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\Big) = a\Big(-\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\Big)$ [lies outside C]

④ $z = ae^{i7\pi/4} = a\Big(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}\Big)$

$= a\Big(\cos\big(2\pi - \frac{\pi}{4}\big) + i\sin\big(2\pi - \frac{\pi}{4}\big)\Big)$

$= a\Big(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\Big) = a\Big(\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\Big)$ [lies outside C]

Hence $z = ae^{i\pi/4}$ and $z = ae^{i3\pi/4}$ are the only simple poles lying inside C.

Hence ($z = ae^{i\pi/4}$, simple pole)

$$\begin{aligned} \Big(\operatorname{Res} f(z)\Big)_{z=ae^{i\pi/4}} &= \lim_{z\to ae^{i\pi/4}} \big(z - ae^{i\pi/4}\big)\frac{z^2}{z^4 + a^4} \\ &= \frac{0}{0} \text{ form} \qquad (\because\ ae^{i\pi/4} \text{ is a root of } z^4 + a^4) \end{aligned}$$

$\Rightarrow$ Applying L'Hospital's rule,

$$\begin{aligned} \Big(\operatorname{Res} f(z)\Big)_{z=ae^{i\pi/4}} &= \lim_{z\to ae^{i\pi/4}} \left[\frac{\big(z - ae^{i\pi/4}\big)(2z) + z^2(1)}{4z^3}\right] \\ &= \frac{0 + \big(ae^{i\pi/4}\big)^2}{4\big(ae^{i\pi/4}\big)^3} \\ \Big(\operatorname{Res} f(z)\Big)_{z=ae^{i\pi/4}} &= \frac{1}{4ae^{i\pi/4}} \end{aligned} \tag{i}$$

Now ($z = ae^{i3\pi/4}$, simple pole)

$$\Big(\operatorname{Res} f(z)\Big)_{z=ae^{i3\pi/4}} = \lim_{z\to ae^{i3\pi/4}} \big(z - ae^{i3\pi/4}\big)\frac{z^2}{z^4 + a^4}$$

Applying L'Hospital's rule, we get

$$= \frac{\big(ae^{i3\pi/4}\big)^2}{4\big(ae^{i3\pi/4}\big)^3}$$

$$\Big(\text{Res } f(z)\Big)_{z=ae^{i3\pi/4}} = \frac{1}{4ae^{i3\pi/4}} \tag{ii}$$

Substituting (i) and (ii) in (1), we get

$$\begin{aligned} I &= \pi i \left[\Big(\text{Res } f(z)\Big)_{z=ae^{i\pi/4}} + \Big(\text{Res } f(z)\Big)_{z=ae^{i3\pi/4}}\right] \\ &= \pi i \left[\frac{1}{4ae^{i\pi/4}} + \frac{1}{4ae^{i3\pi/4}}\right] \\ &= \frac{\pi i}{4a}\left[\frac{1}{\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}} + \frac{1}{\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}}\right] \end{aligned}$$

$$I = \frac{\pi i}{4a}\left[\frac{1}{\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}} + \frac{1}{-\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}}\right]$$

$$= \frac{\pi i}{4a}\left[\frac{-\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}+\frac{i}{\sqrt{2}}}{\left(i\frac{1}{\sqrt{2}}\right)^2-\left(\frac{1}{\sqrt{2}}\right)^2}\right]$$

$$I = \frac{\pi i}{4a}\left[\frac{\frac{2i}{\sqrt{2}}}{-\frac{1}{2}-\frac{1}{2}}\right] = \frac{-\pi\, 2/\sqrt{2}}{4a(-1)} \qquad (\because\ i\cdot i = -1)$$

$$\boxed{I = \frac{\pi}{2a\sqrt{2}}}$$

Exercises (Type II)

Exercises (Type II)

1. Show that $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{x^4+1} = \frac{\pi}{\sqrt{2}}$.
2. Show that $\displaystyle\int_0^{\infty} \frac{x^2}{(1+x^2)^2}\,dx = \frac{\pi}{4}$.
3. Show that $\displaystyle\int_0^{\infty} \frac{x^2}{(x^2+1)(x^2+4)}\,dx = \frac{\pi}{6}$.